Introduction to Digital Image Processing

Unit- 1

One picture is worth more than ten thousands words.

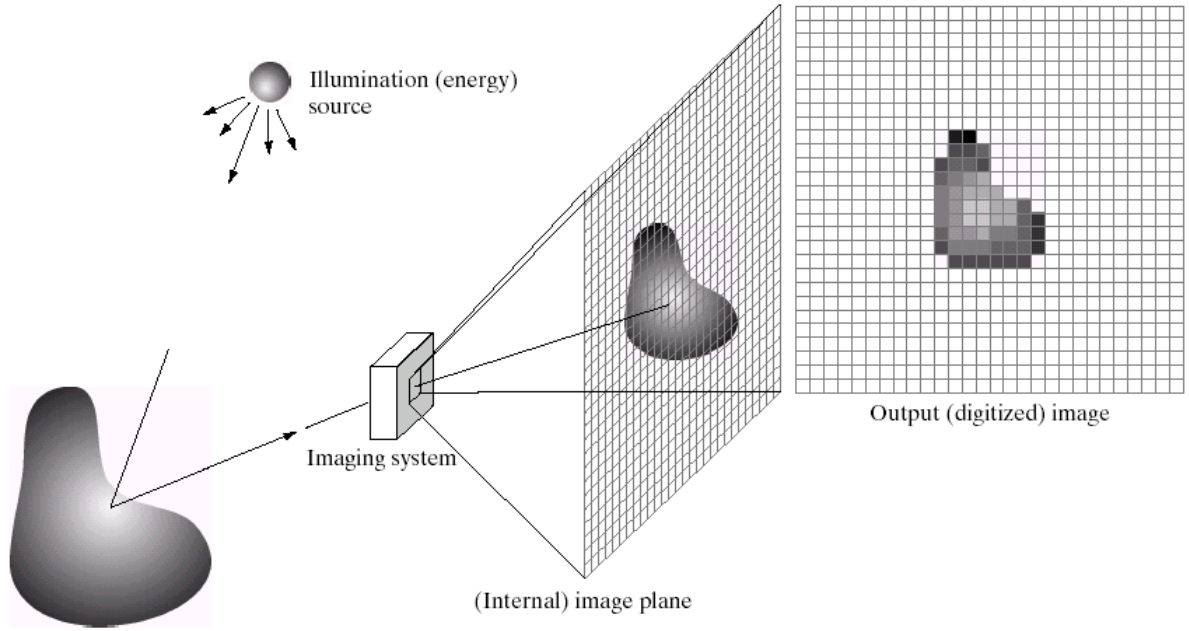
-Anonymous

What is Image?

- An image is a 2D visual representation of an object, a person, or a scene produced by an optical device such as a mirror, a lens, or a camera.
- It may be defined as two-dimensional function, f(x,y), where x and y are spatial (plane) coordinates and the amplitude of f at any pair of coordinates (x,y) is called intensity or gray level of the image at that point.

What is a Digital Image?

- A digital Image is a representation of a two-dimensional image using a finite number of points, usually referred to as picture elements (which is also referred to as image elements, pels or pixels).
- Each pixel is represented by one or more numerical values:



Digital Image Representation

- A digital image is an image f(x,y) that has been discretized both in spatial co-ordinates and in brightness.
- It is represented by 2D array. The digitized brightness value is called gray level values.
- Digital image can be represented as a matrix of M x N size as shown in figure below where x denotes the row number from 0 to M-1 and y represents the column number from 0 to N-1:

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0, N-1) \\ f(1,0) & f(1,1) & \cdots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1, N-1) \end{bmatrix}$$

What is Digital Image Processing (DIP)?

- The term digital image processing generally refers to processing/manipulation/analysis of a two dimensional picture, usally by a electronic means or digital computer by means of digitization (sampling and quantization).
- Shortly, processing a digital image to get required information out of it or to make it the way user want it using computer or any other electronic means is called digital image processing.

Goal

- Digital image processing focuses on two major tasks:
  - Improvement of pictorial information for human interpretation
  - Processing of image data for storage, transmission and representation for autonomous machine perception

Some argument about where image processing ends and fields such as image analysis and computer vision start

Three levels of image processing operations

Low Level Process	Mid Level Process	High Level Process
Input: Image **Output:** Image	**Input:** Image **Output:** Attributes	**Input:** Attributes **Output:** Understanding
Examples: Noise removal, image sharpening	**Examples:** Object recognition, segmentation	**Examples:** Scene understanding, autonomous navigation

Example of Typical Image Processing Operations

1. Sharpening
2. Noise Removal
3. Deblurring
4. Edge extraction
5. Binarization
6. Blurring
7. Contrast Enhancement
8. Object Segmentation and Lablling

1. Sharpening

- A technique by which the edge and fine details of an image are enhanced for human viewing.
- This is done in the spatial domain as well as in frequency-domain.

(a) (b)

2. Noise Removal

- Image processing filters can be used to reduce the amount of noise in an image before processing it any further.
- Depending on the type of noise, different noise removal techniques are used

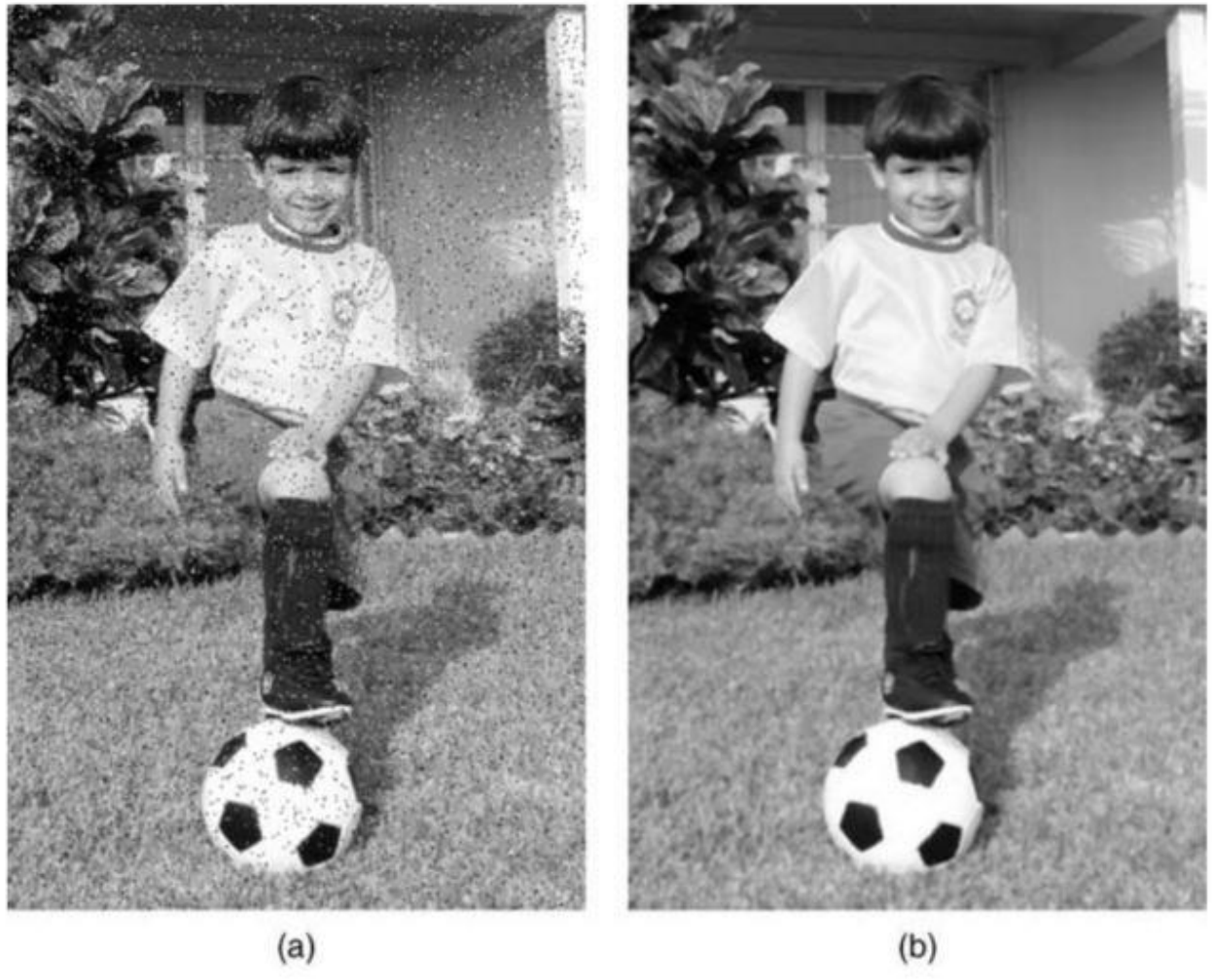


Figure: Deblurring: (a) Original Image; (b) after removing the noise

3. Deblurring

- An image may appear blurred for many reasons, ranging from improper focusing of the lens to an insufficient shutter speed for a fast-moving object.

(a) (b)

Figure: Deblurring: (a) Original Image; (b) after removing the (motion) blur

4. Edge Extraction

- Extracting edges from an image is a fundamental preprocessing step used to separate objects from one another before identifying their contents.

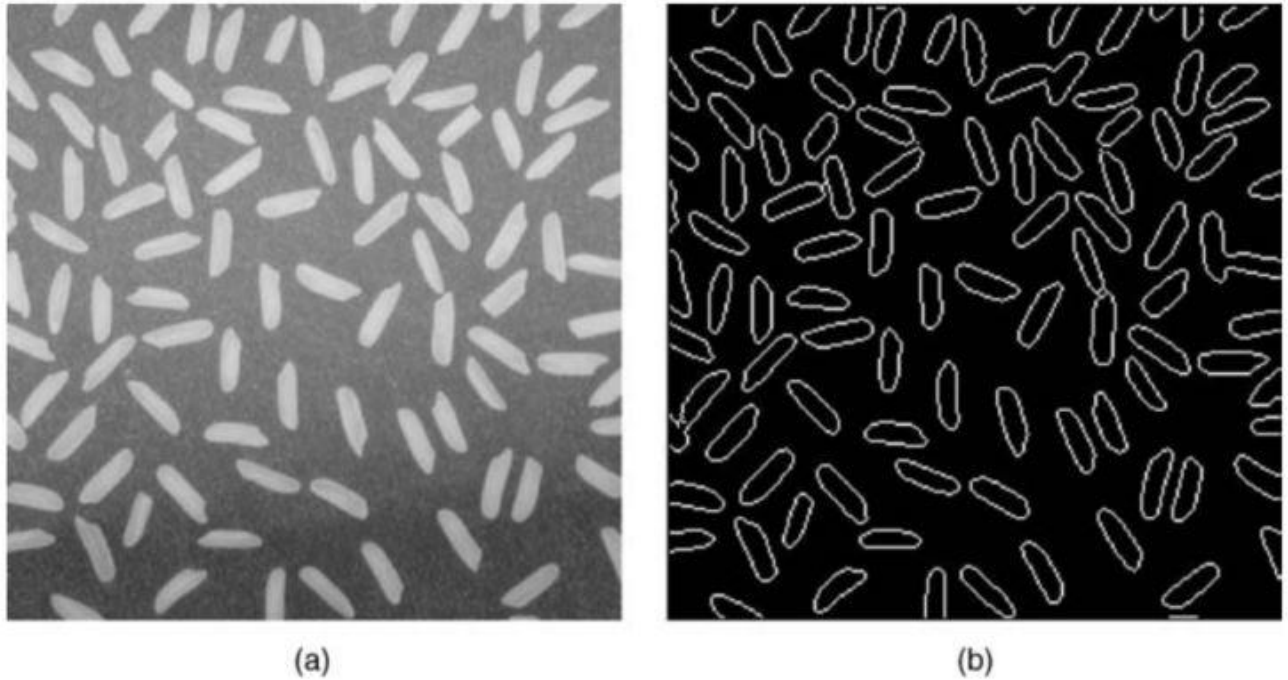


Figure: Edge Extraction: (a) Original Image; (b) after extracting its most relevant edges

5. Binarization

- In many image analysis applications, it is often necessary to reduce the number of gray levels in a monochrome image to simplify and speed up its interpretation.
- Reducing a grayscale image to only two levels of gray (black and white) is usually referred to as *binarization*.

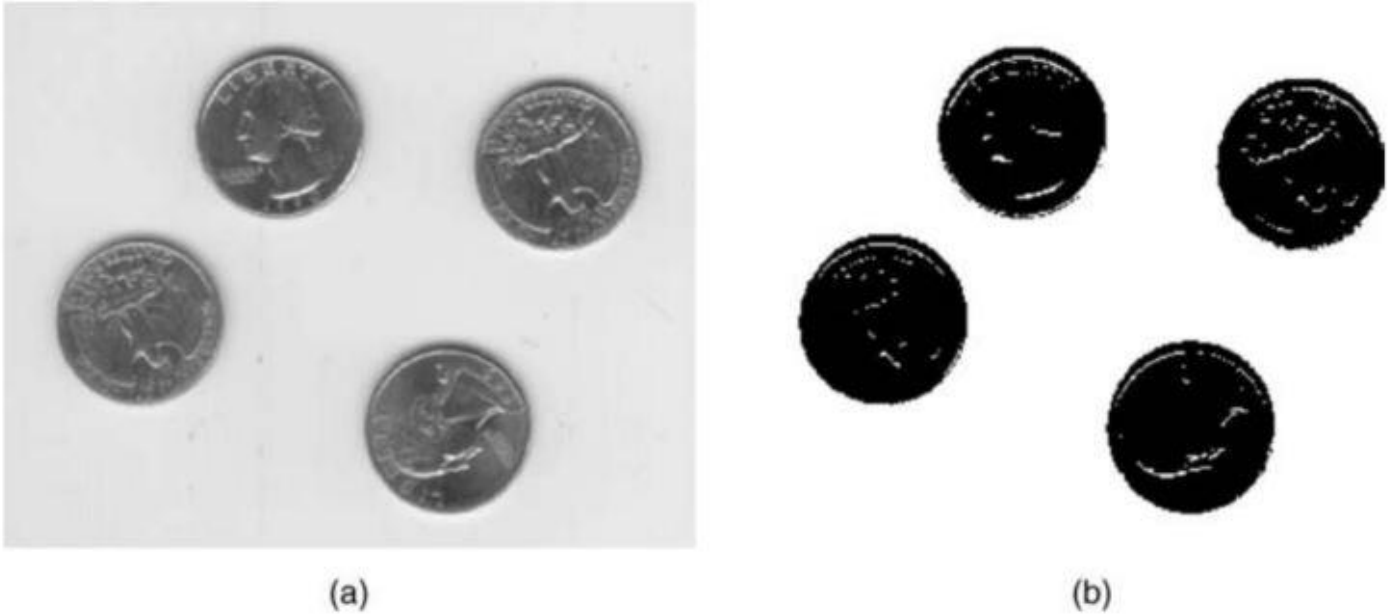
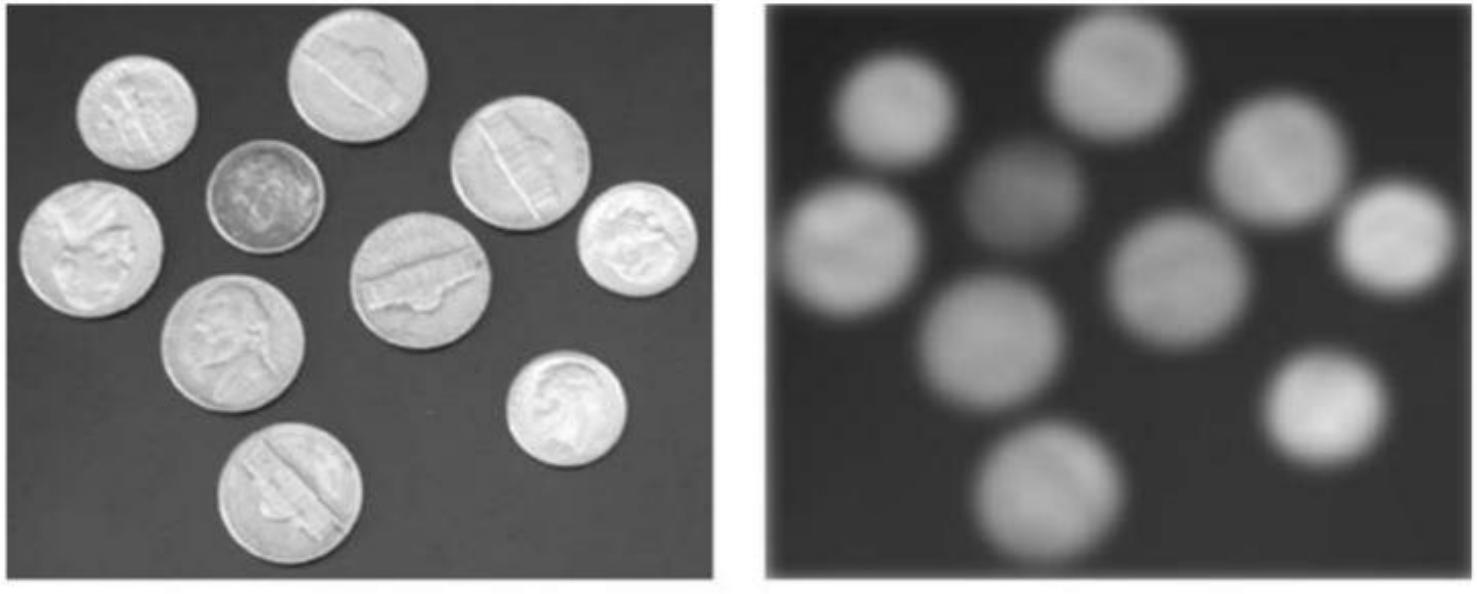


Figure: Binarization: (a) Original Image; (b) after conversion to a black-and-white version

6. Blurring

- It is sometimes necessary to blur an image in order to minimize the importance of texture and fine details in a scene, for instance, in cases where objects can be better recognized by their shape.



(a) (b)

7. Contrast Enhancement

- In order to improve an image for human viewing as well as make other image processing tasks (e.g. edge extraction) easier, it is often necessary to enhance the contrast of an image.
- Contrast enhancement techniques are used by transformation functions and histogram processing.

(a) (b)

Figure Contrast Enhancement (a) Original Image (b) after histogram equalization to increase contrast

8. Segmentation and Labeling

- The task of segmenting and labeling objects within a scene is a prerequisite for most objects recognition and classification systems.
- Once the relevant objects have been segmented and labeled, their relevant features can be extracted and used to classify, compare, cluster, or recognize the objects in question.

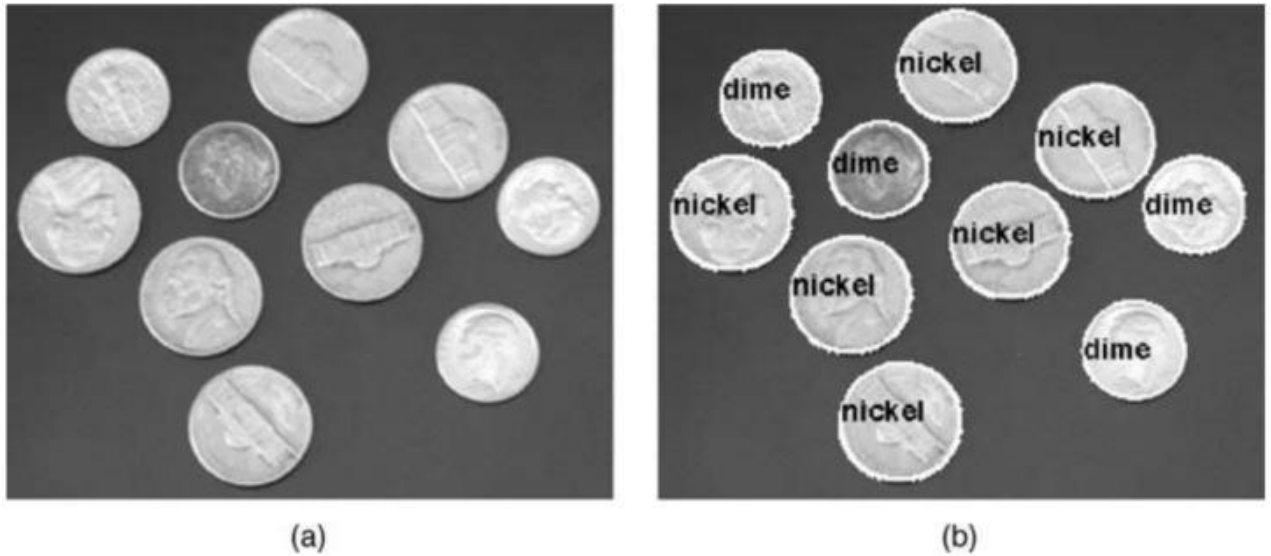


Figure: Object Segmentation and Labeling: (a) Original Image; (b) after segmenting and labeling

Fundamental Steps in Digital Image Processing

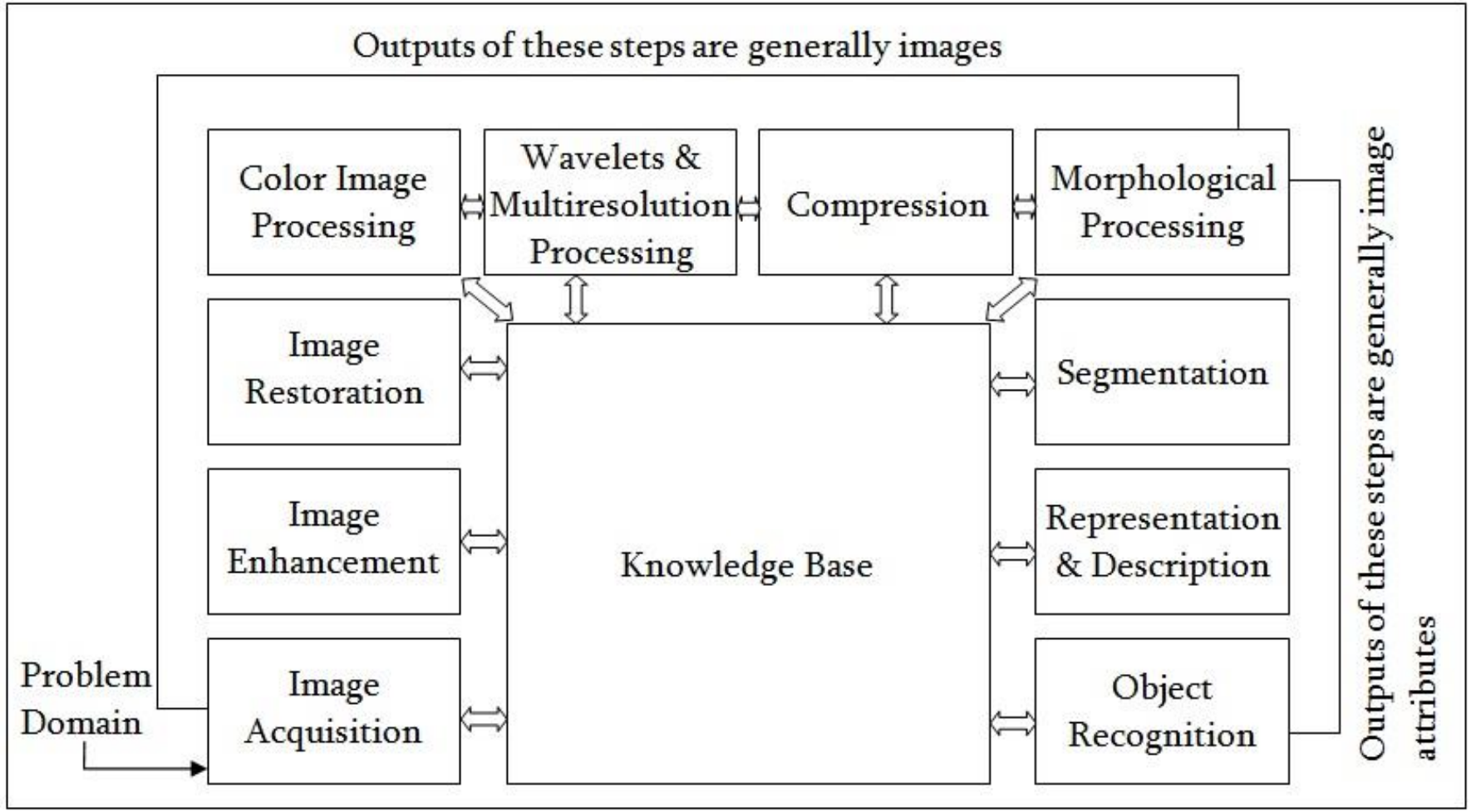


Figure: Fundamental Steps in Digital Image Processing

1. Image Acquisition

- This is the first step or process of the fundamental steps of digital image processing.
- Image acquisition could be as simple as being given an image that is already in digital form.
- Generally, the image acquisition stage involves pre-processing, such as scaling etc.

2. Image Enhancement

- Image enhancement is among the simplest and most appealing areas of digital image processing.
- It is the process of manipulating an image so that the result is more suitable than the original for a specific application.
- Basically, the idea behind enhancement techniques is to bring out detail that is obscured, or simply to highlight certain features of interest in an image such as, changing brightness & contrast etc.

3. Image Restoration

- Image restoration is an area that also deals with improving the appearance of an image.
- However, unlike enhancement, which is subjective, image restoration is objective, in the sense that restoration techniques tend to be based on mathematical or probabilistic models of image degradation.
- Enhancement, on the other hand, is based on human subjective preferences regarding what constitutes a “good” enhancement result.

4. Color Image Processing

- Color image processing is an area that has been gaining its importance because of the significant increase in the use of digital images over the Internet.
- This may include color modeling and processing in a digital domain etc.

5. Wavelets and Multi-Resolution Processing

- Wavelets are the foundation for representing images in various degrees of resolution.
- Images subdivision successively into smaller regions for data compression and for pyramidal representation.

6. Compression

- Compression deals with techniques for reducing the storage required to save an image or the bandwidth to transmit it.
- Particularly in the uses of internet it is very much necessary to compress data.

7. Morphological Processing

- Morphological processing deals with tools for extracting image components that are useful in the representation and description of shape.

8. Segmentation

- Segmentation procedures partition an image into its constituent parts or objects.
- In general, autonomous segmentation is one of the most difficult tasks in digital image processing.
- A rugged segmentation procedure brings the process a long way toward successful solution of imaging problems that require objects to be identified individually.

9. Representation and Description

- Representation and description almost always follow the output of a segmentation stage, which usually is raw pixel data, constituting either the boundary of a region or all the points in the region itself.
- Choosing a representation is only part of the solution for transforming raw data into a form suitable for subsequent computer processing.
- Boundary representation is appropriate when the focus in on external shape characteristics, such as corners and inflections.
- Regional representation is appropriate when the focus is on internal properties, such as texture or skeletal shape.
- *Description*, also called *feature selection*, deals with extracting attributes that result in some quantitative information of interest or are basic for differentiating one class of objects from another.

10. Object Recognition

- Recognition is the process that assigns a label, such as, “vehicle” to an object based on its descriptors.

11. Knowledge Base

- Knowledge about a problem domain is coded into an image processing system in the form of a knowledge database.
- This knowledge may be as simple as detailing regions of an image where the information of interest is known to be located, thus limiting the search that has to be conducted in seeking that information.
- The knowledge base also can be quite complex, such as an interrelated list of all major possible defects in a materials inspection problem or an image database containing high-resolution satellite images of a region in connection with change-detection applications.
- The knowledge base also controls the interaction between the processing modules.

Components of Image Processing

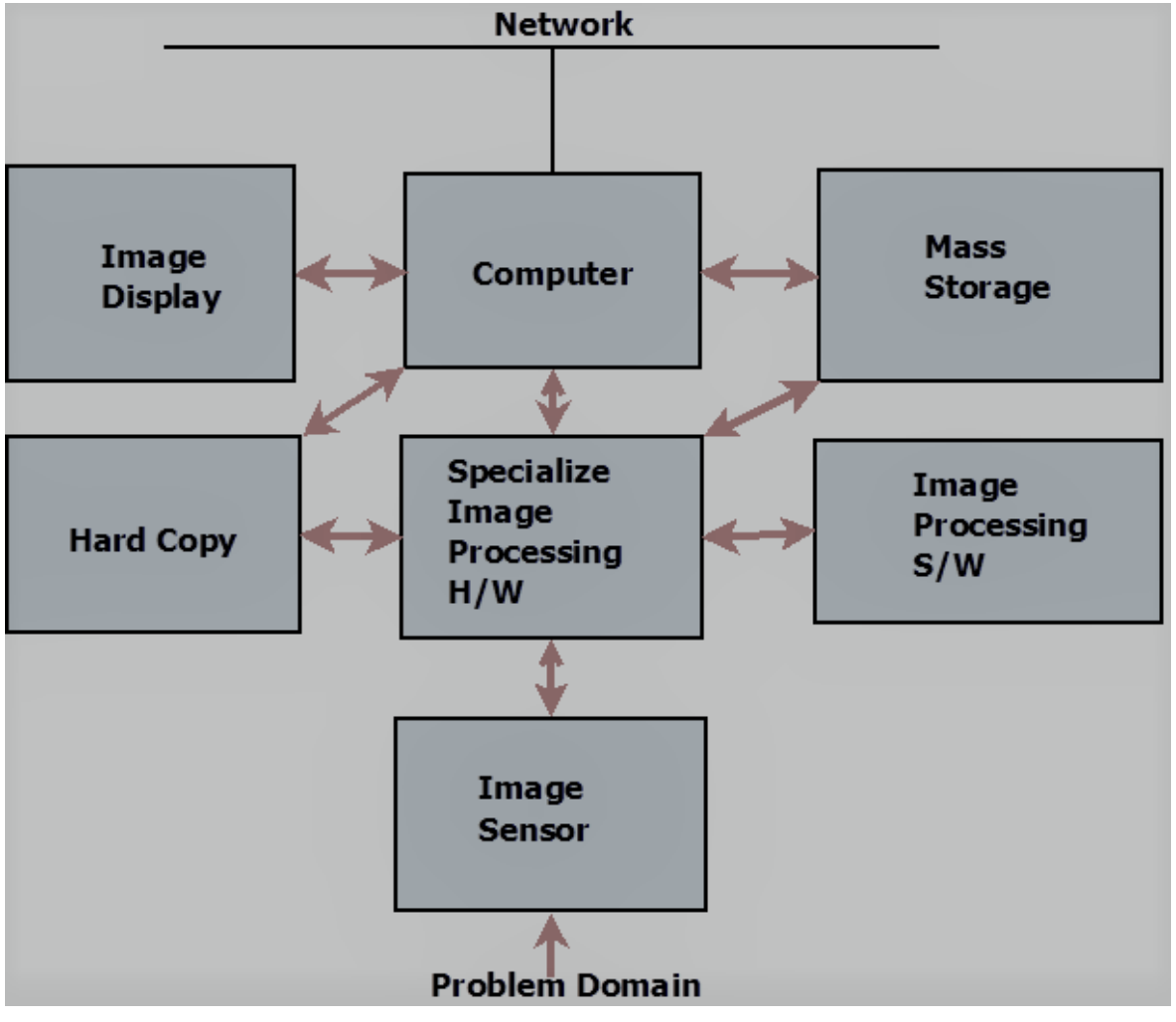


Figure: Components of a general-purpose image processing system

1. Image Sensor

- With reference to sensing, two elements are required to acquire digital images.
- The first is a physical device that is sensitive to the energy radiated by the object we wish to image such as camera, scanner etc.
- The second, called a ***digitizer***, is a device for converting the output of the physical sensing device into digital form i.e. it produce an image by the digitization technique.

2. Specialized Image Processing Hardware

- It usually consists of the digitizer just mentioned, plus hardware that performs other primitive operations, such as an arithmetic logic unit (ALU), which performs arithmetic and logical operations in parallel on entire images.

- One example of how an ALU is used is in averaging images as quickly as they are digitized, for the purpose of noise reduction.

- This type of hardware sometimes is called a *front-end subsystem*, and its most distinguishing characteristic is speed. In other words, this unit performs functions that require fast data throughputs (e.g., digitizing and averaging video images at 30 frames) that the typical main computer cannot handle.

3. Computer

- The computer is image processing is a general purpose computer and can range from a PC to a supercomputer
- Almost, it is suitable for off-line image processing tasks.

4. Image Processing Software

- Software in image processing consists of specialized module that performs specific task. It also includes the capability for the user to write code that utilizes the specialize modules

5. Mass Storage

- Mass storage is must in image processing applications.
- It is used to store the image line by line.
- Storage in measured in bytes.
- Digital storage for image processing application falls into three principal categories:
  1. Short-term storage for use during processing; e.g. Frame buffers
  2. On-line storage for relatively fast recall; e.g. magnetic disks and optical-media storage
  3. Archival storage, characterized by infrequent access; e.g. magnetic tapes and optical disks

6. Image Displays

- Image display are TV monitors used to display images.
- Monitors are driven by the outputs of image and graphics display cards that are an integral part of the computer system.

7. Hardcopy

- Hardcopy devices for recording images include laser printers, film cameras, heat-sensitive device, inkjet units, and digital units such as optical and CD-ROM disks.
- For presentation of images, image projection equipment's are used.

8. Networking

- Networking is almost a default function in any computer system in use today.
- Because of the large amount of data inherent in image processing applications, the key consideration in image transmission is bandwidth.
- Optical fiber and other broadband technologies are use in practice for connecting remote sites via Internet.

Electromagnetic Spectrum

- The principal energy source for images in use today is the electromagnetic energy spectrum. Other important sources of energy include acoustic, ultrasonic, and electronic (in the form of electron beams used in electron microscopy).
- Images based on radiation from the EM spectrum are the most familiar especially images in the X-ray and visual bands of the spectrum.

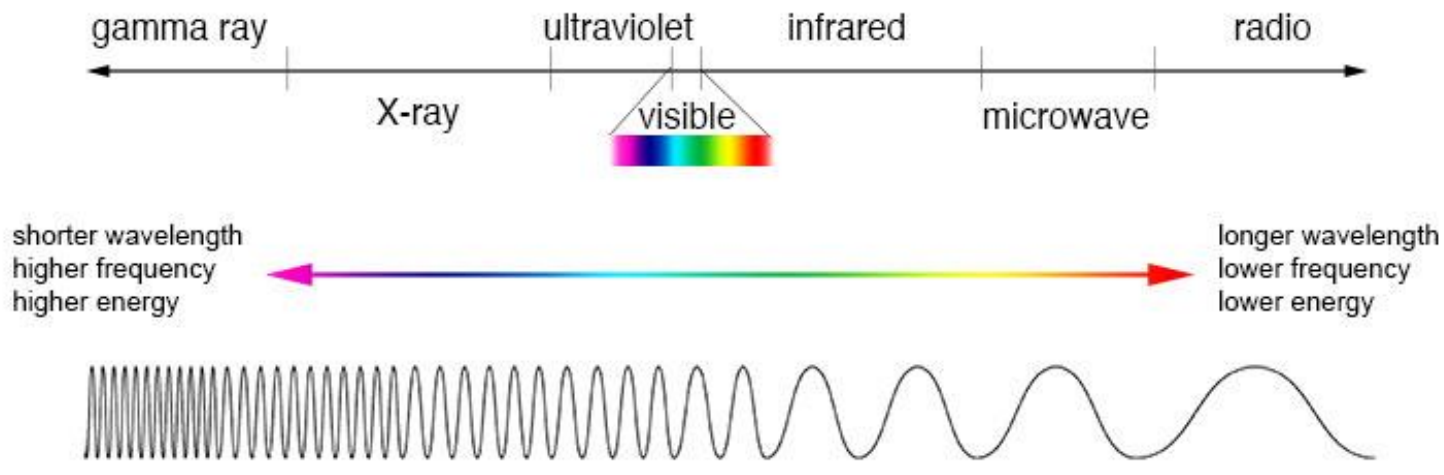


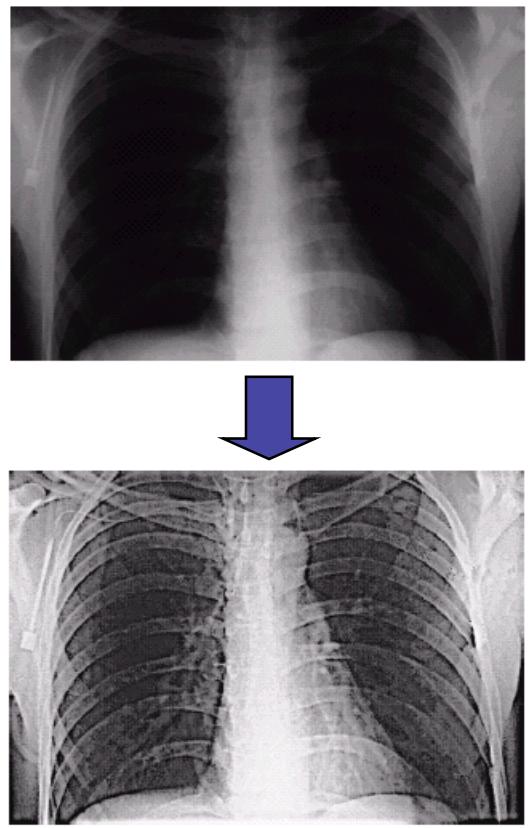
Figure: The electromagnetic Spectrum

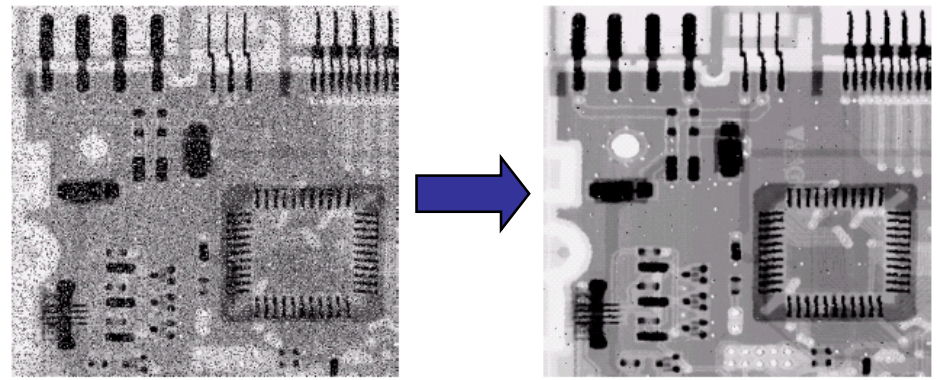
Application of Image Processing

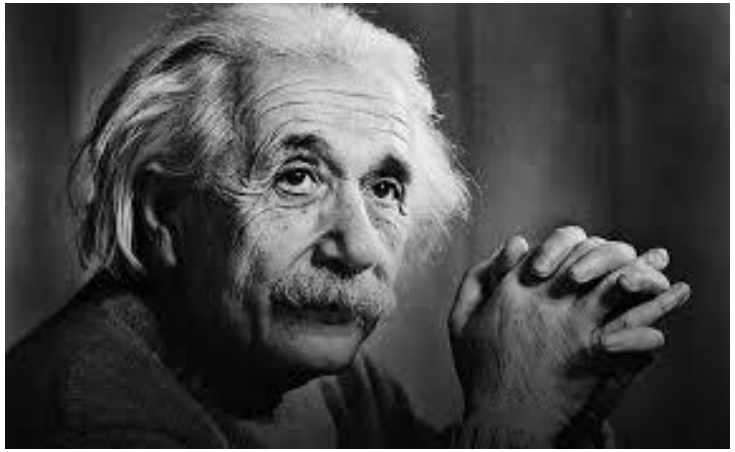
- Image Enhancement and restoration
- Artistic effects
- Medical visualisation
- Industrial inspection
- Law enforcement
- Human computer interfaces

1. Image Enhancement and restoration

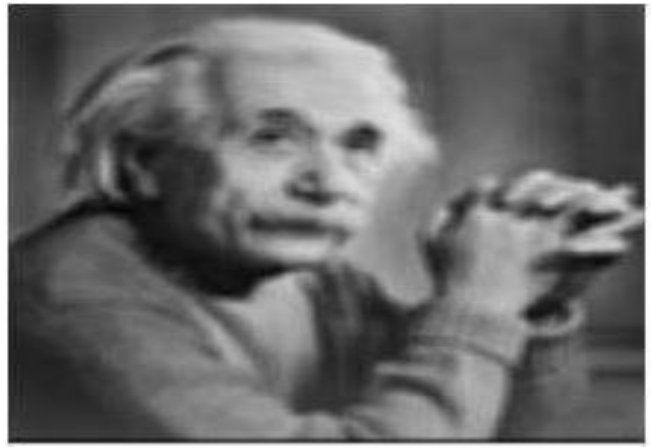
- Image sharpening and restoration refers here to process images that have been captured from the modern camera to make them a better image or to manipulate those images in way to achieve desired result.
- This includes removing noise, Zooming, blurring , sharpening , gray scale to color conversion, detecting edges and vice versa , Image retrieval and Image recognition.



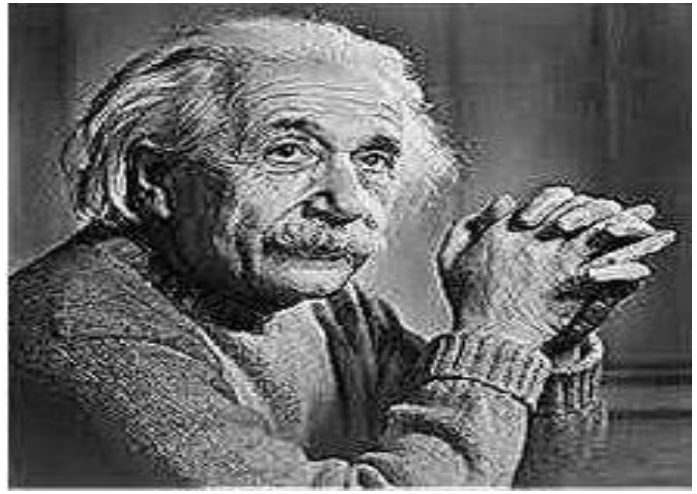




Original Image



Blurred Image



Sharp Image

2. Artistic Effects

Artistic effects are used to make images more visually appealing, to add special effects and to make composite images

3. Medical Visualization

- The major applications in the filed of medical are:
  - Gamma Ray Imaging
    - Used to locate bone pathology, such as infections or tumors
    - Positron Emission tomography (PET)
  - X-Ray Imaging
  - Medical CT
  - UV imaging

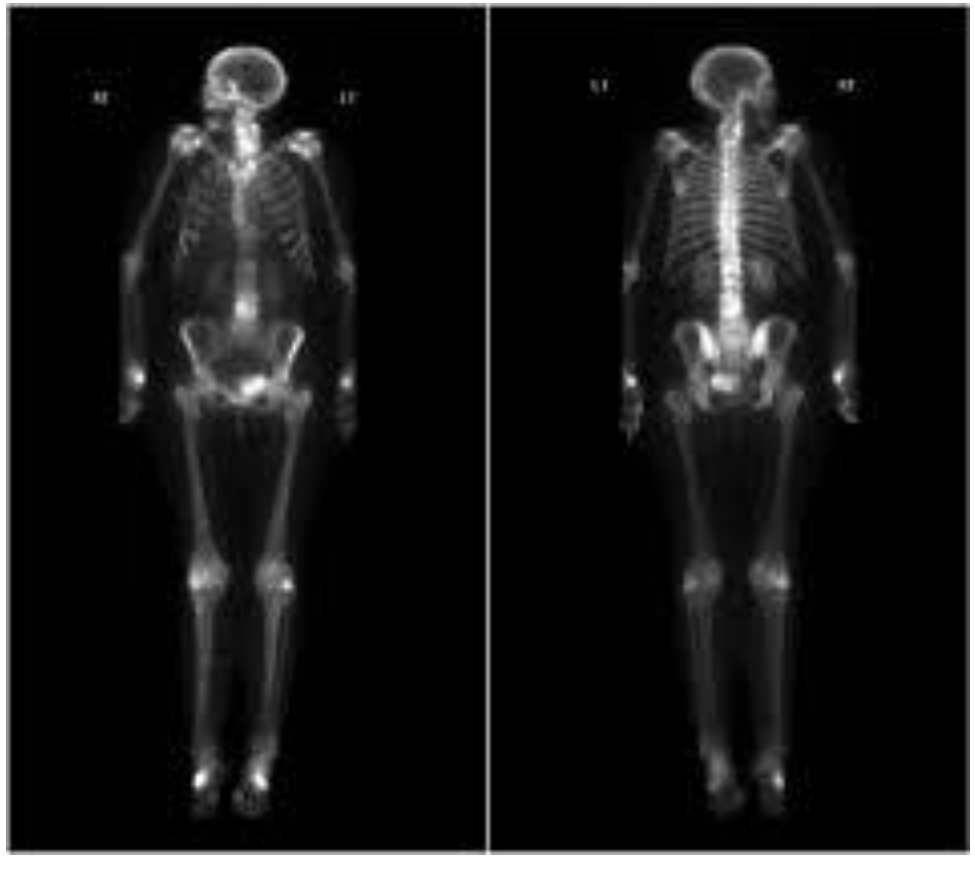


Figure: Gamma Ray Imaging – Bone Scan

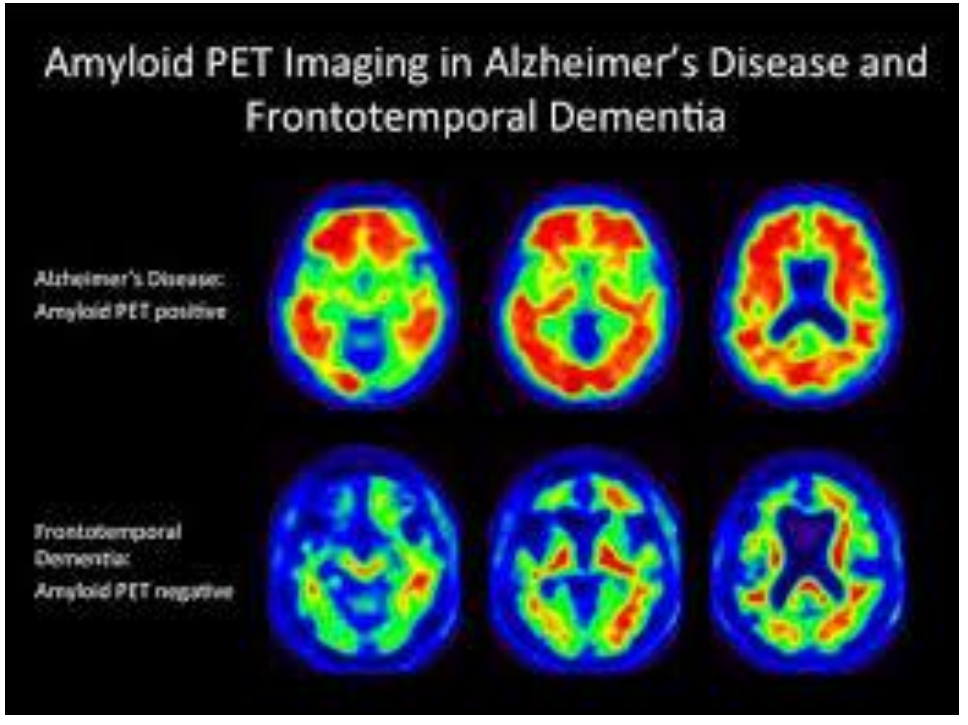


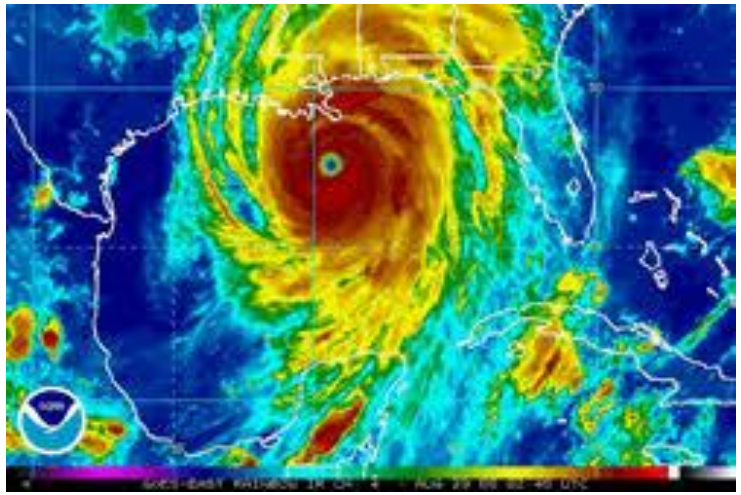
Figure: PET Imaging

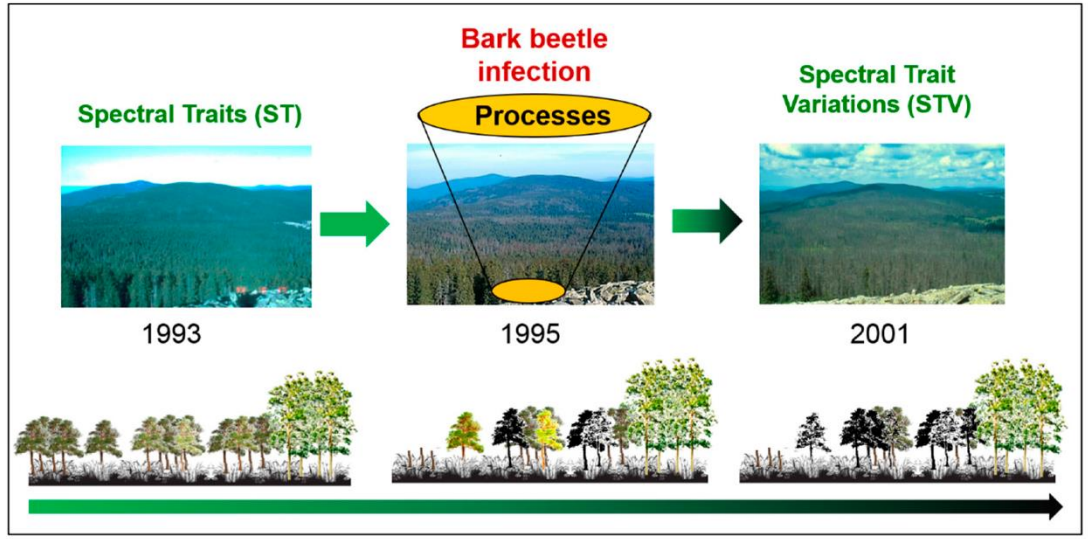
Figure: Angiogram Imaging

4. GIS

Geographic Information Systems

- Digital image processing techniques are used extensively to manipulate satellite imagery
- Terrain classification
- Meteorology





Night-Time Lights of the World data set

- Global inventory of human settlement
- Not hard to imagine the kind of analysis that might be done using this data

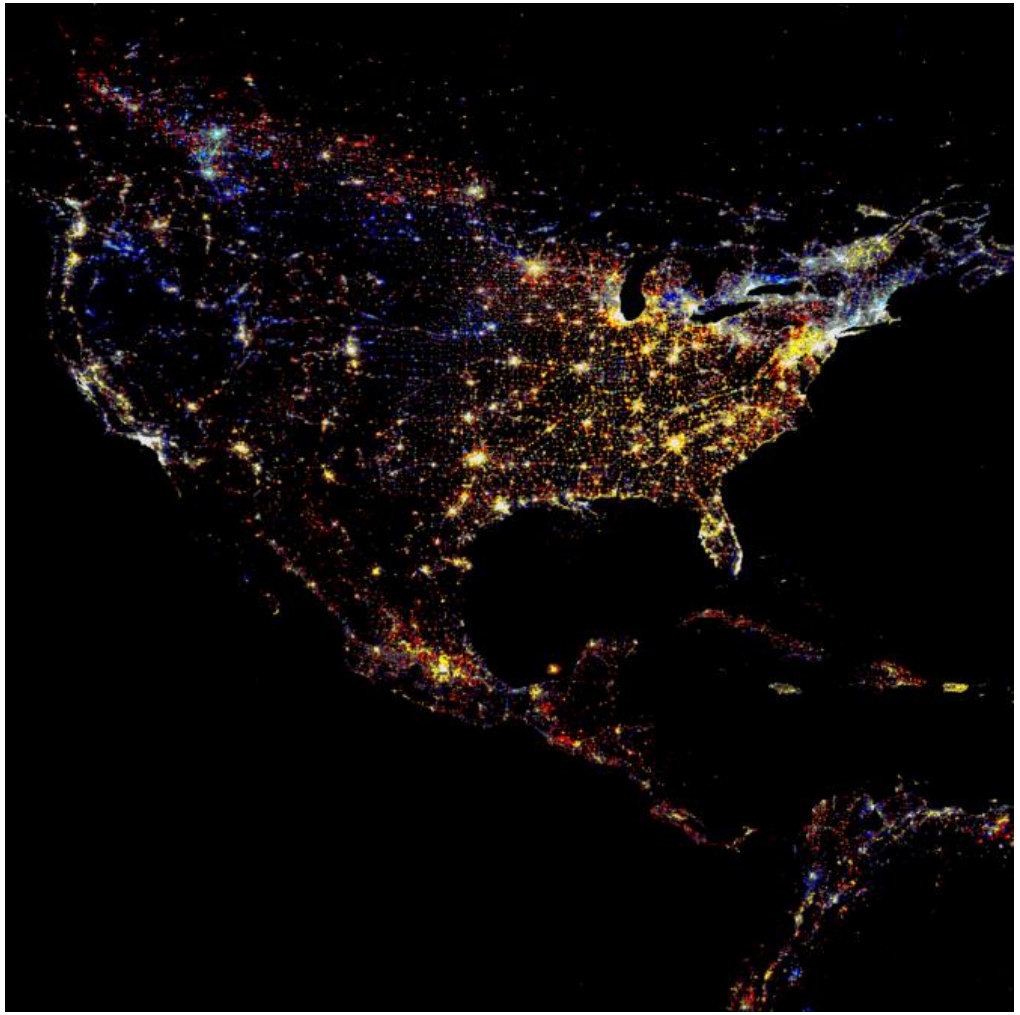
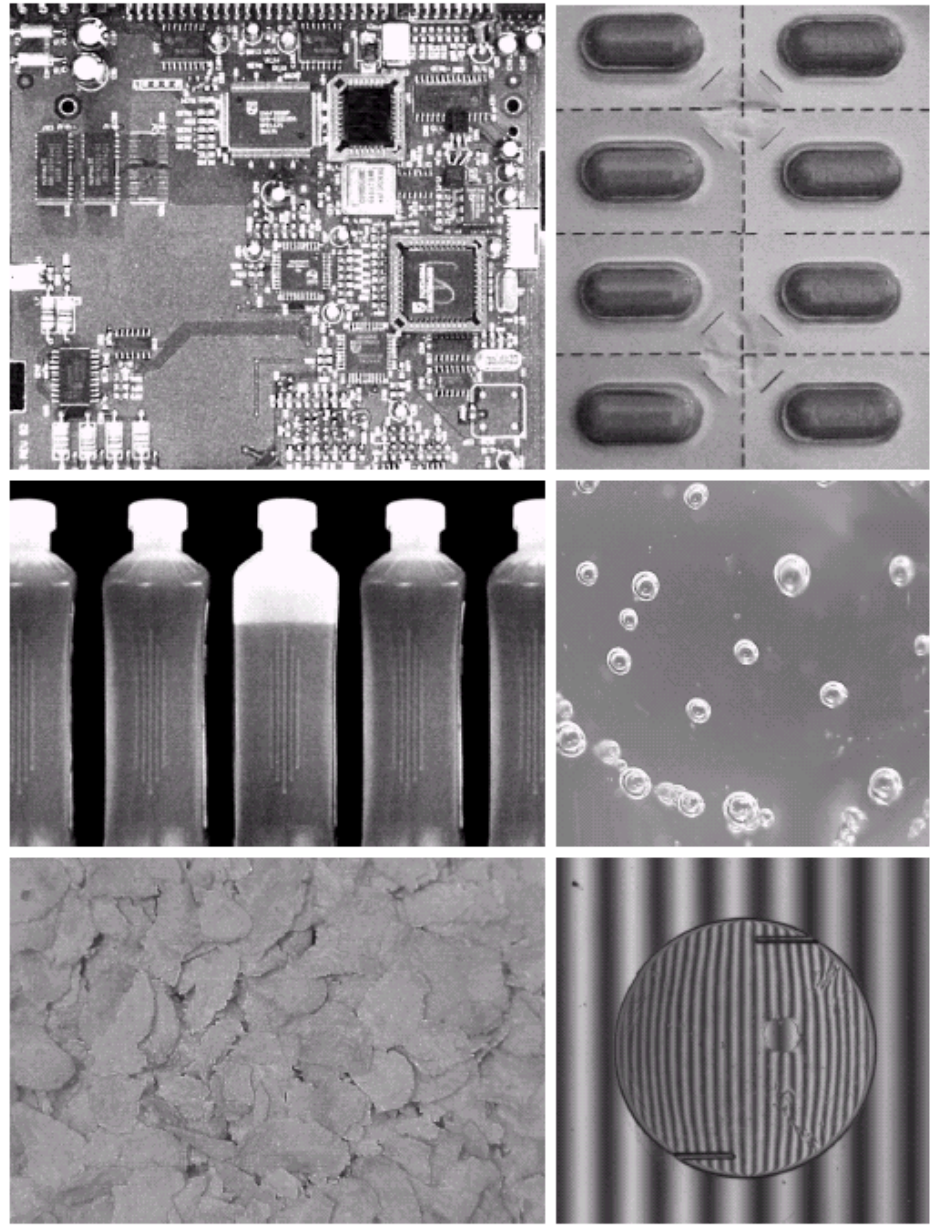


Figure: Night-Time Lights of the World

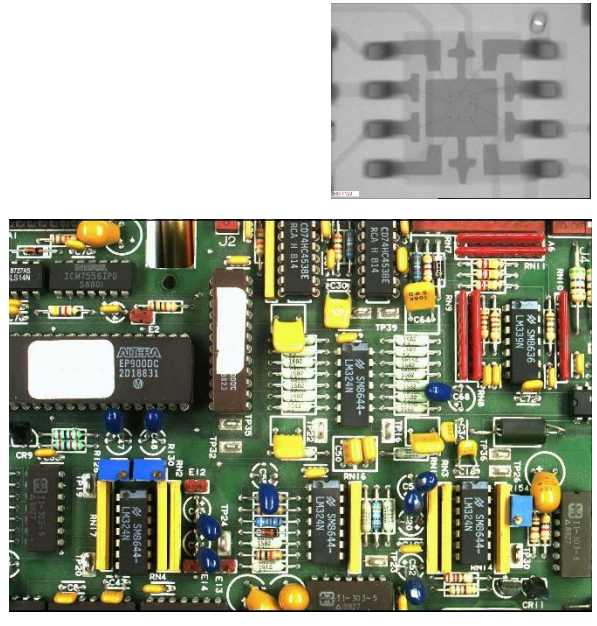
5. Industrial Inspection

- Human operators are expensive, slow and unreliable
- Make machines do the job instead
- Industrial vision systems are used in all kinds of industries
- Can we trust them?



Printed Circuit Board (PCB) inspection

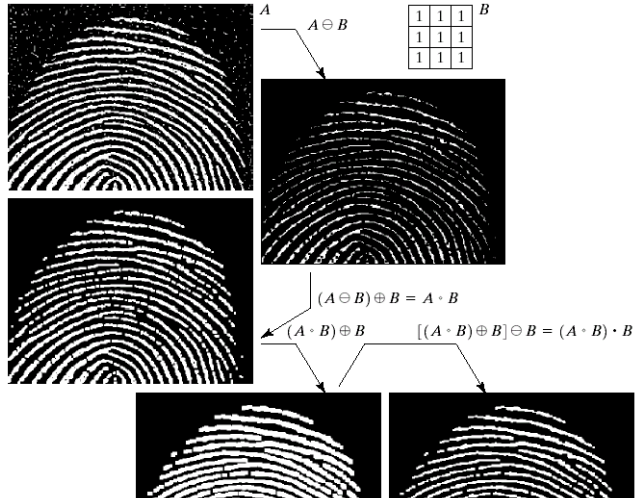
- Machine inspection is used to determine that all components are present and that all solder joints are acceptable
- Both conventional imaging and x-ray imaging are used



6. Law Enforcement

Image processing techniques are used extensively by law enforcers

- Number plate recognition for speed cameras/automated toll systems
- Fingerprint recognition
- Enhancement of CCTV images



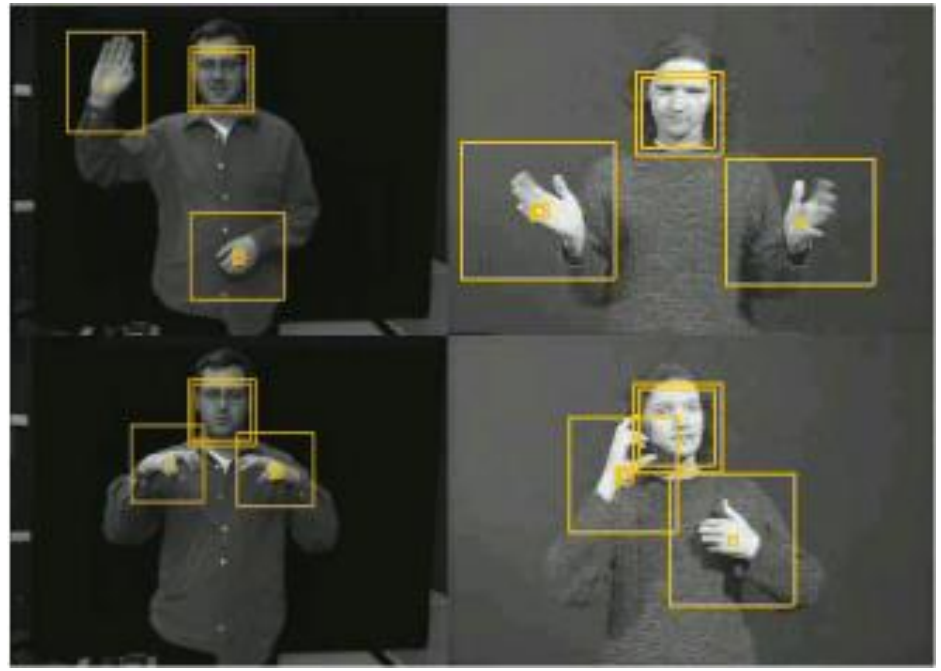
7. HCI

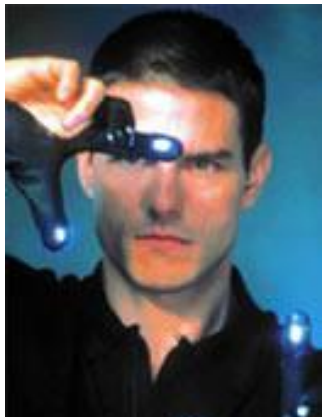
Try to make human computer interfaces more natural

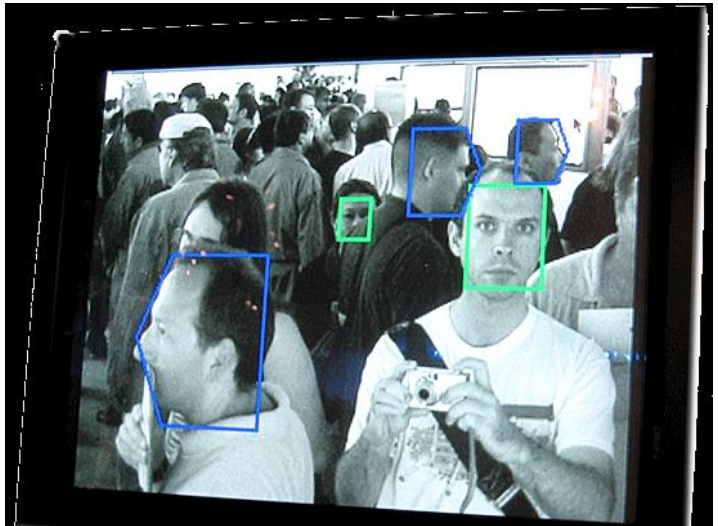
- Face recognition
- Gesture recognition

Does anyone remember the user interface from “Minority Report”?

These tasks can be extremely difficult







Elements of Visual Perception

Human visual perception

- Knowledge of how images form in the eye can help us with processing digital images.

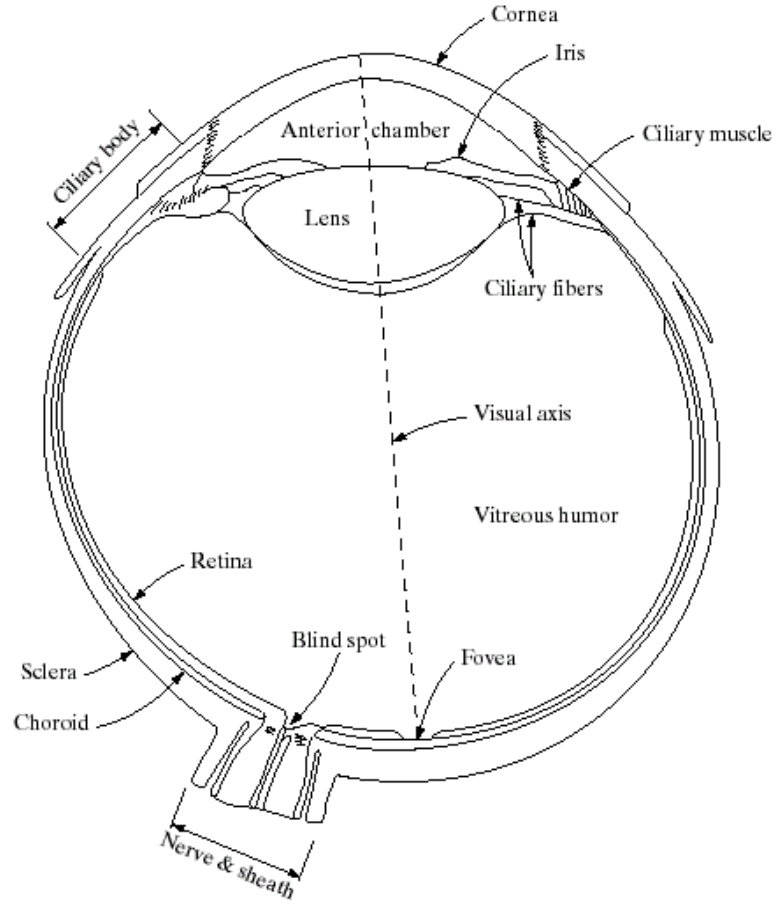


Figure: Structure of Human Eye

Structure of the Human Eye

- The eye is nearly sphere, with an average diameter of approximately 20 mm.
- Three membranes enclose the eye: the *cornea* and *sclera* outer cover; the *choroid*, and the *retina*.
- The choroid is divided into the *ciliary body* and the *iris*.
- The iris controls the amount of light that enters into the eye. It is 2 to 8 mm in diameter.
- The lens focuses light from objects onto the retina.
- The retina is covered with light receptors called cones (6-7 million) and rods (75-150 million).
- Cones are concentrated around the fovea and are very sensitive to color. Cone vision is *photopic* or *bright-light vision.*
- Rods are more spread out retina surface and are sensitive to low levels of illumination.

Image Formation in the Eye

- In an ordinary photographic camera, the lens has fixed focal length, and the focusing at various distances is achieved by varying the distance between the lens and the imaging plane, where the film is located.
- In the human eye, the converse is true; the distance between the lens and the imaging region (the retina) is fixed, and the focal length needed to achieve the proper focus is obtained by varying the shape of the lens.

- Muscles within the eye can be used to change the shape of the lens allowing us focus on objects that are near or far away.
- An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain.

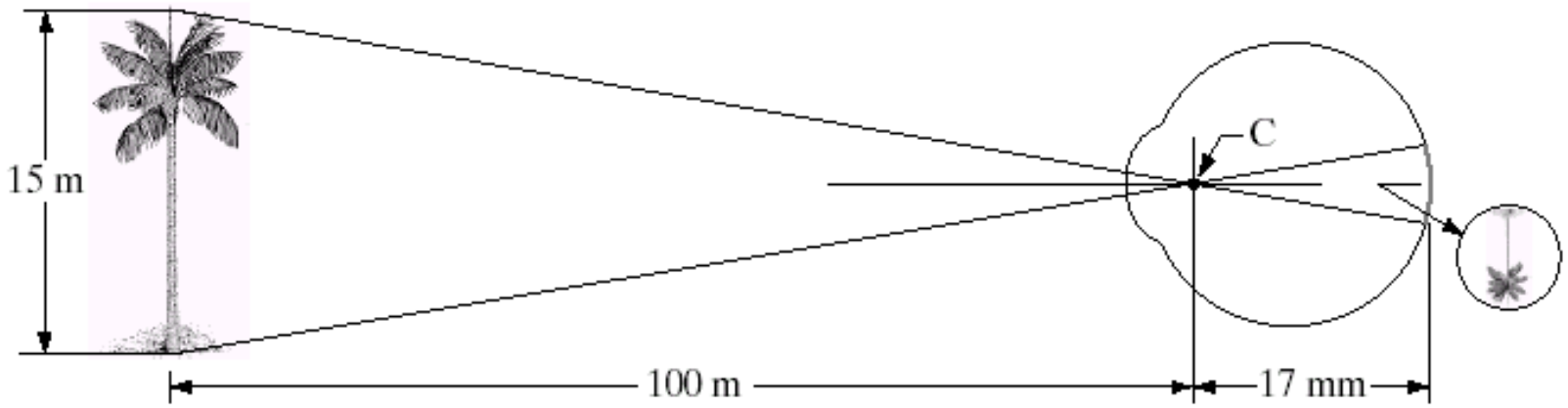


Figure: Graphical representation of the eye looking at the palm tree. Point C is the optical center of the lens.

The figure illustrate how to obtain the dimensions of an image formed on the retina. For example, suppose a person is looking at a tree 15 m high at the a distance 100m. Letting h denote the height of that object in the retinal image, the geometry of figure above yields 15/100 = h/17 or h = 2.55 mm. therefore, the height of retrieval image is

Brightness Adaptation and Discrimination

- The human visual system can perceive approximately 10^{10} different light intensity levels.
- However, at any one time we can only discriminate between a much smaller number – *brightness adaptation.*
- Similarly, the *perceived intensity* of a region is related to the light intensities of the regions surrounding it.

Figure: An example of Mach Band.

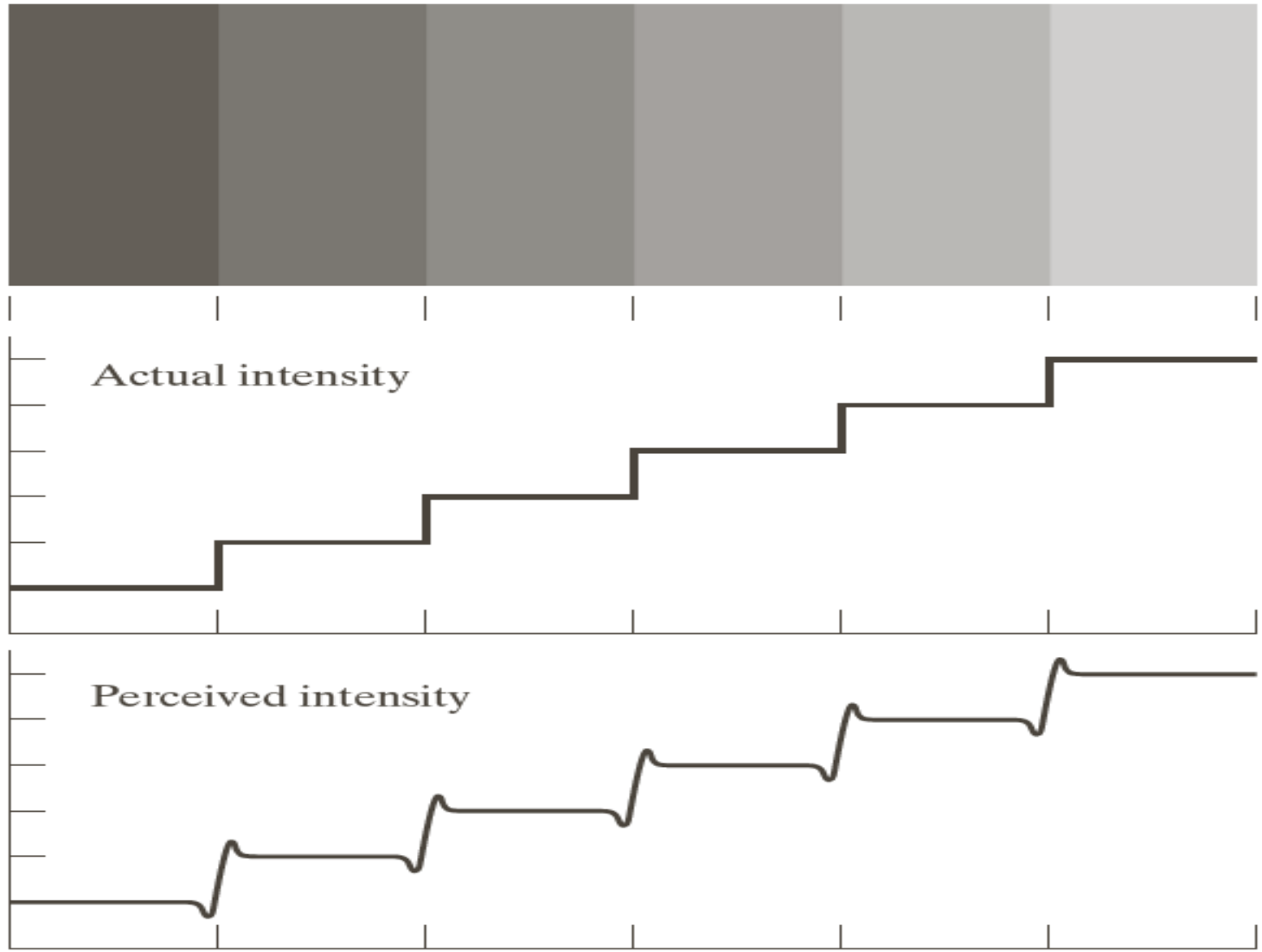


Figure: Illustration of the Mach Band effect. Perceived intensity is not a simple function of actual intensity.

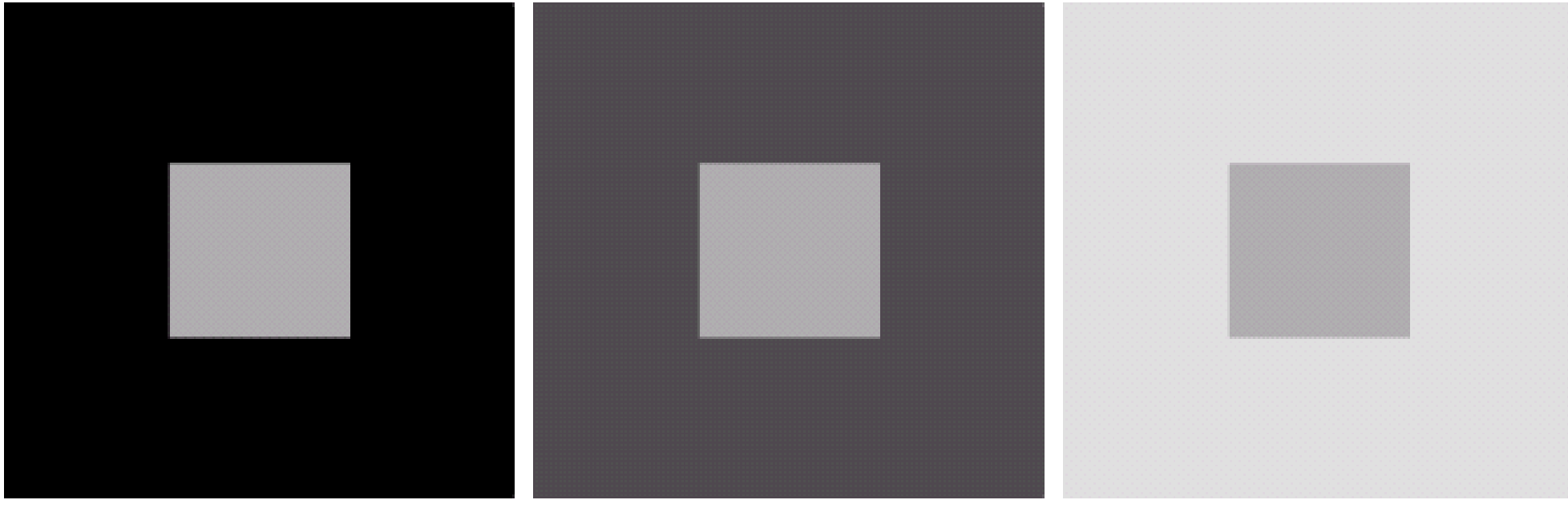
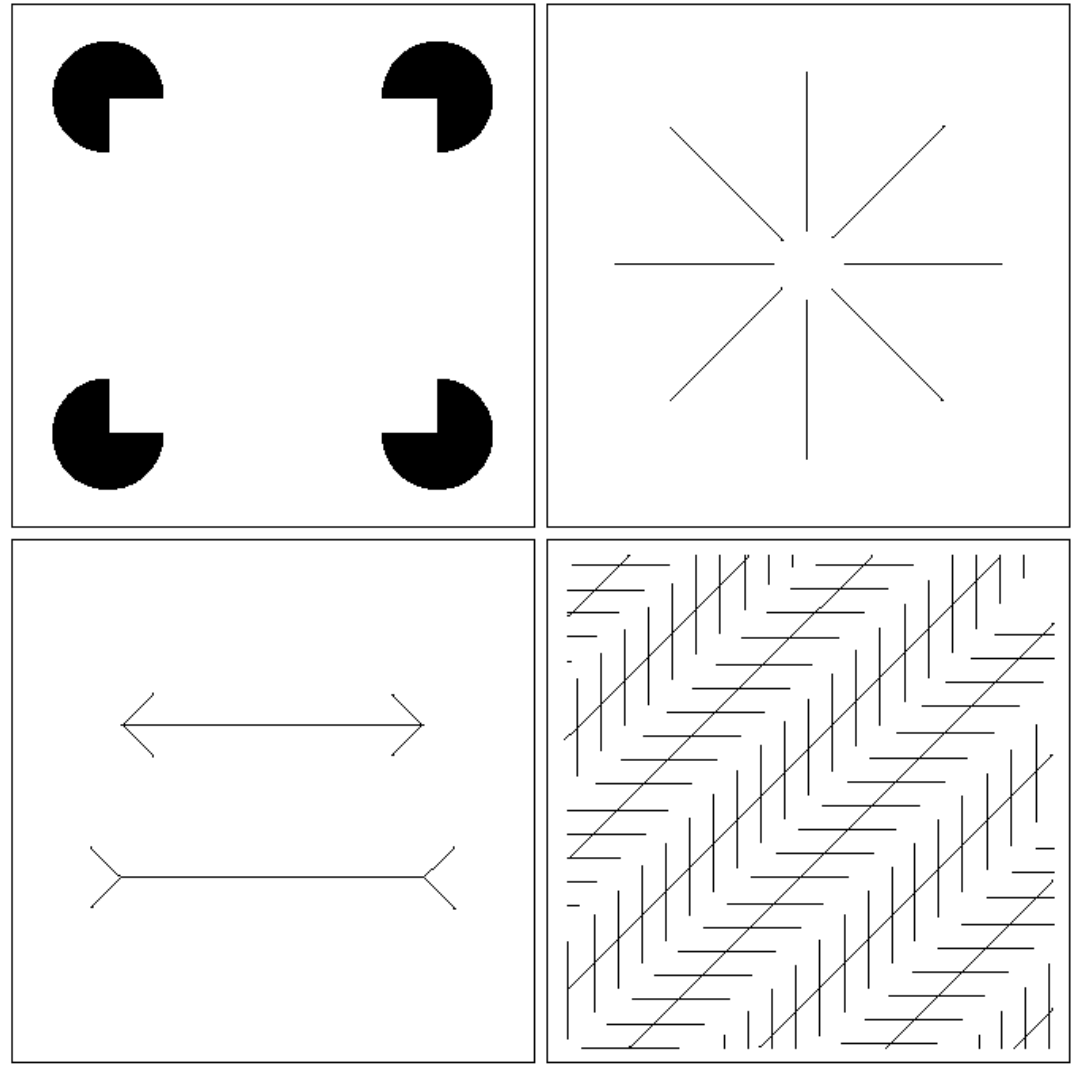


Figure: An example of *simultaneous contrast.* All the inner square have the same intensity, but they appear progressively darker as the background becomes lighter.

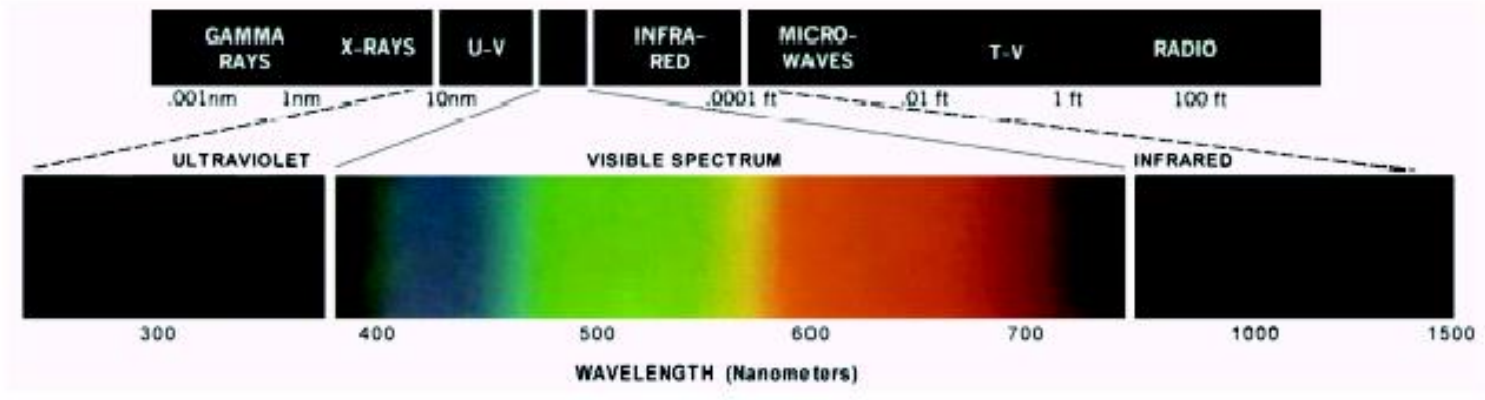
Optical Illusion

- Our visual systems play lots of interesting tricks on us.
- In the first figure, the outline of a square is seen clearly, despite the fact that no lines defining such a figure are part of the image.
- In the second figure, a few lines give the illusion of a circle.
- In the third figure, the two horizontal lines are of same length, but one appears shorter than the other.
- In the last figure, all lines oriented at 450 are equidistance and parallel; but the crosshatching creates the illusion that those lines are far from being parallel.



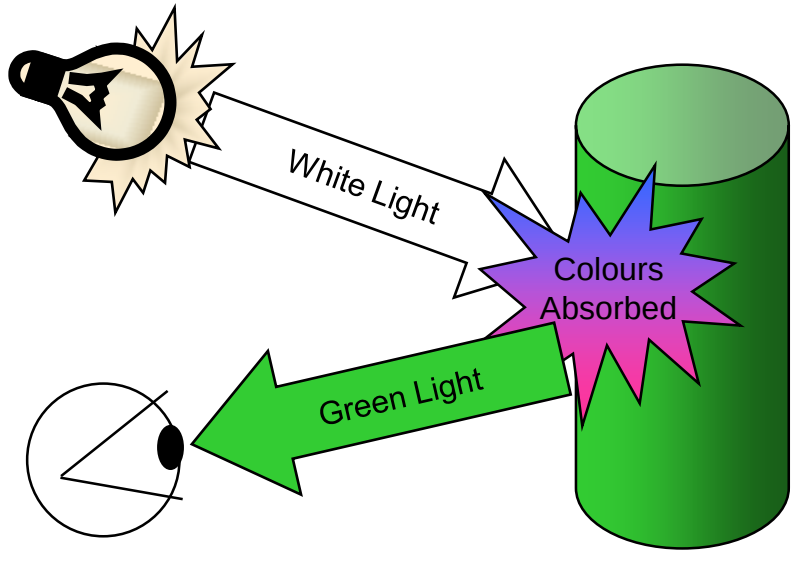
Light and the Electromagnetic Spectrum

- Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye.
- The electromagnetic spectrum is split up according to the wavelengths of different forms of energy.



Reflected Light

- The color that we perceive are determined by the nature of the light from an object.
- For example, if light is shone onto a green object most wavelength are absorbed, while green light is reflected from the object. (Green light have wavelength primarily in the range 500 nm to 570 nm).



Light Types

- Monochromatic Light / Achromatic Light
- Chromatic Light / Color Light

Monochromatic or Achromatic light

- *Light that is void of color*.
- The only attribute of monochromatic light is its *intensity* or amount.
- The intensity of monochromatic light is perceived to vary from black to gray and finally to white, so the term *gray level* is used to denote the monochromatic intensity.
- The range of measured values of monochromatic light from black to white is usually called the *gray scale*.
- The monochromatic images are called *gray-scale images*.

Chromatic or Color Light

- The electromagnetic spectrum from approximately 0.43 to 0.79 μm.
- Three basic quantities are used to describe the quality of a chromatic light source:

1. **Radiance**
   - It is the total amount of energy that flows from the light source.
   - Measure in watts (W).
2. **Luminance**
   - Gives a measure of the amount of energy an observer perceives from light source.
   - Measured in lumens (lm).
   - For example: light emitted from a source operating in the far infrared region of the spectrum could have significance energy (radiance), but an observer would hardly perceive it; its luminance would be almost zero.
3. **Brightness**
   - A subjective description of light perception that is practically impossible to measure

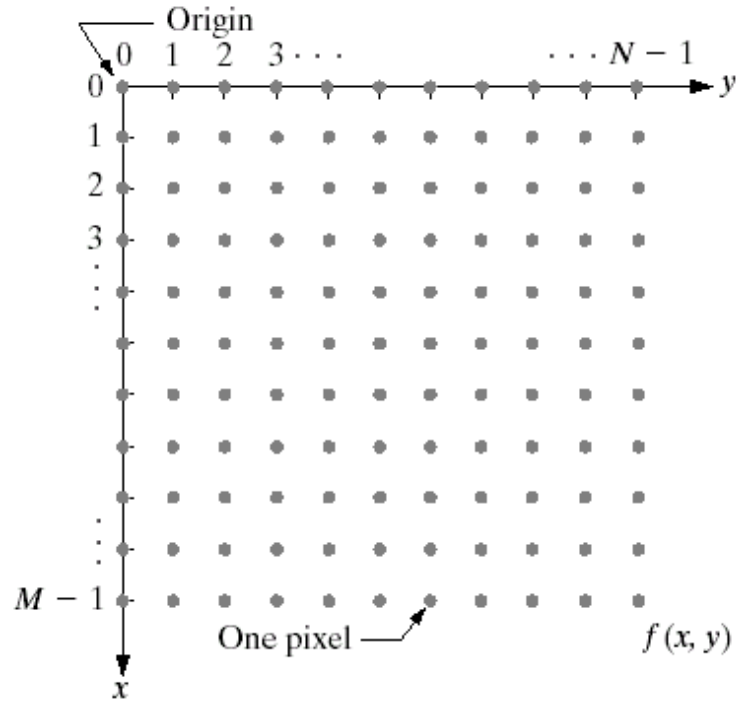
Sampling, Quantization and Resolution

- In the following slides we will consider what is involved in capturing a digital image of a real-world scene
  - ❑ Image sensing and representation
  - ❑ Sampling and quantisation
  - ❑ Resolution

Image Representation

- A digital image is composed of M rows and N columns of pixels each storing a value.
- We denote images by 2D functions of the form *f(x, y)*.
- The value of f at coordinate (x ,y) is a positive scalar quantity, determined by the source of the image.
- When an image is generated by from a physical process, its intensity values are proportional to energy radiated by a physical source (e.g., electromagnetic waves).
- Therefore, the value of f at (x ,y) is

$$0 < f(x,y) < \infty$$



$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

- The fuction f(x,y) is characterized by two components:
  - The amount of the source illumination incident on the scene viewed, called, *illumination component denoted by i(x,y)* and
  - The amount of illumination reflected by the object in the scene, called *reflectance component denoted by r(x,y)*.
- Therefore, $f(x,y) = i(x,y)\, r(x,y)$, where $0 < i(x,y) < \infty$ and $0 < r(x,y) < 1$.

- Let the intensity (gray level) of a monochromatic image at any coordinates (x0,y0) be denoted by $l = f(x_0,y_0)$

  Such that $L_{min} <= l <= L_{max}$

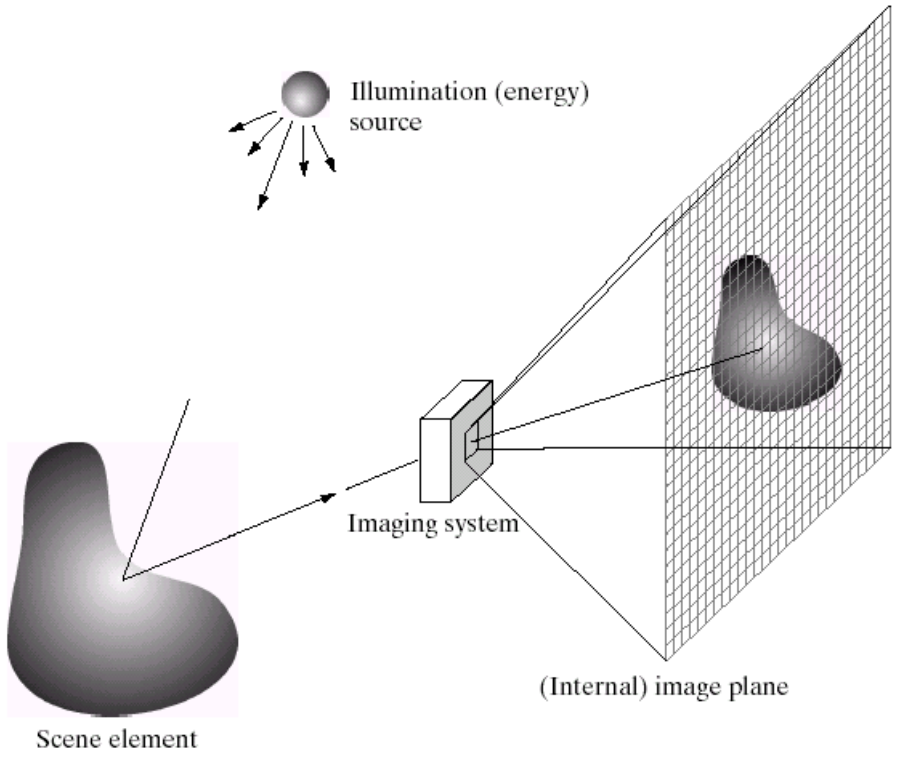
  The interval $[L_{min}, L_{max}]$ is called the <u>gray (or intensity) scale</u>.

Typically, for $[L_{min}, L_{max}] = [0 , L-1]$ where, $l=0$ is considered as black and $l = 255$ is considered as white on the gray scale and the number of intensity levels is an integer power of 2:

$$L = 2^k$$

Image Acquisition

- Images are typically generated by illuminating a scene and absorbing the energy reflected by the objects in that scene.
- Typical notion of illumination and scene can be way off:
  - X-rays of a skeleton
  - Ultrasound of an unborn baby
  - Electro-microscopic images of molecules



Sampling and Quantization

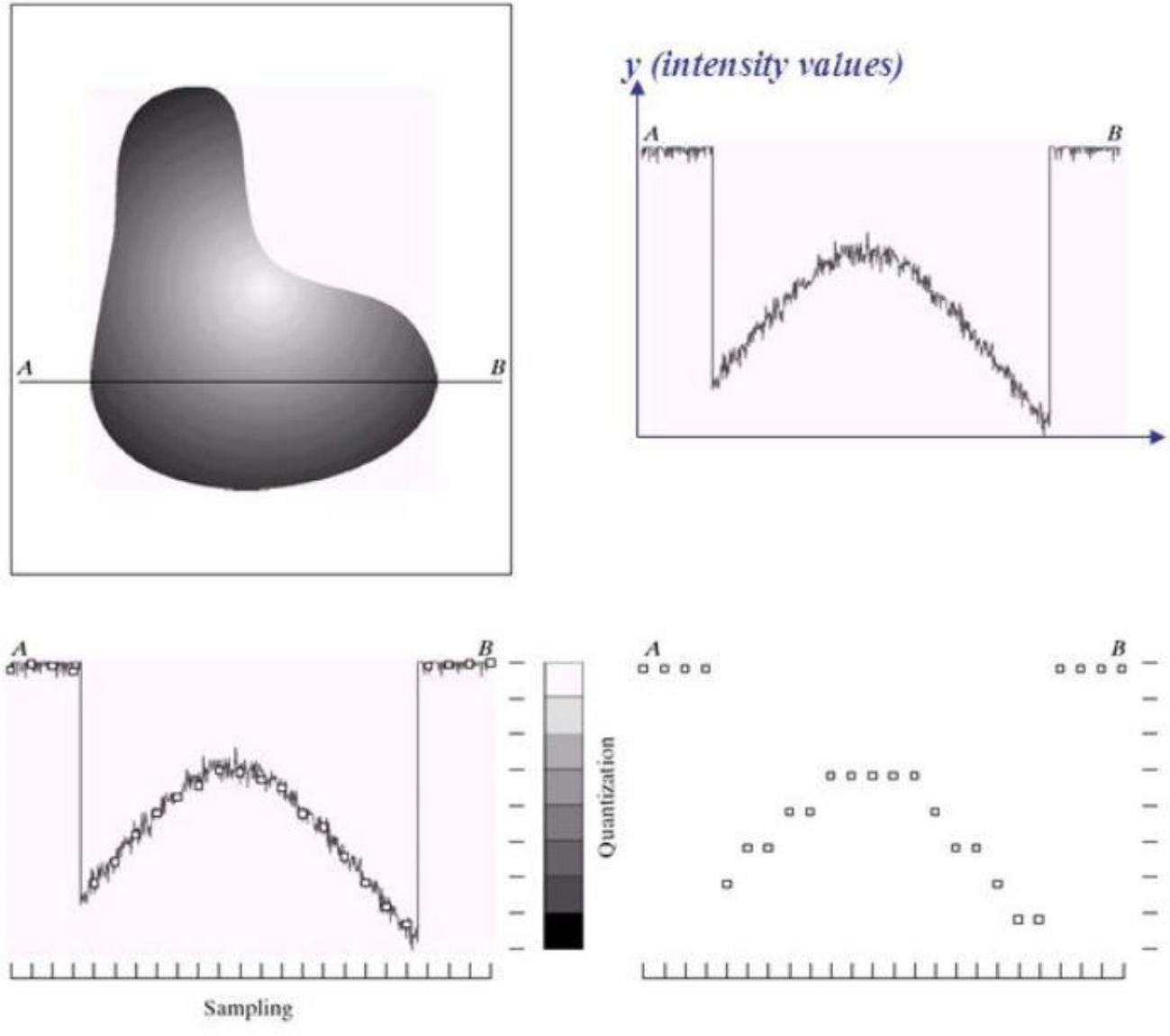
- A physical image is changed into digital image by performing sampling and quantization. This process is known as Digitization.

Sampling

- It is the process where measurement are taken at regular space intervals. *Digitizing the spatial co-ordinates values* (x,y) is called sampling.
- It is the process by which image formed over a patch (domain) in a continuous domain is mapped into a discrete point with integer co-ordinates.

Quantization

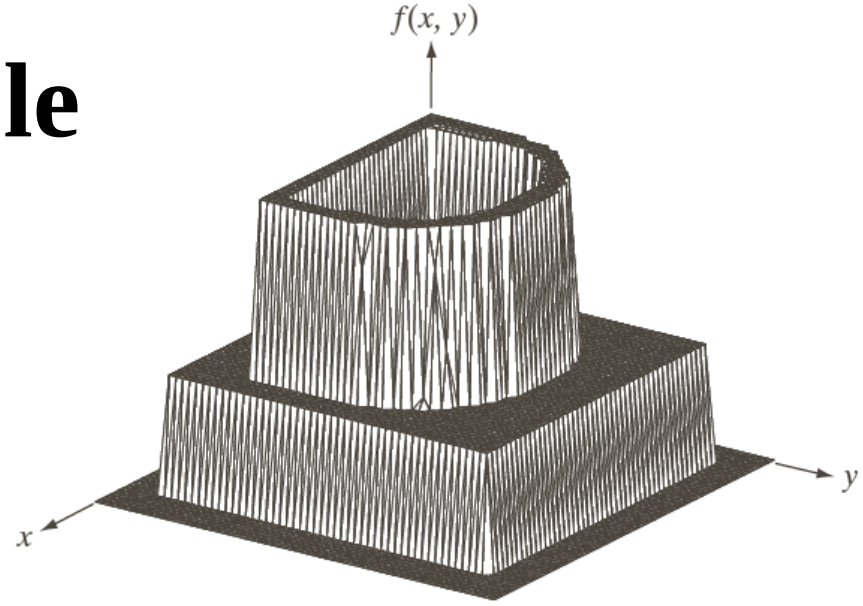
- It is the process of mapping the measured intensity to one of the finite number of discrete levels.
- Hence *digitizing the amplitude values* is called quantization or gray level quantization.

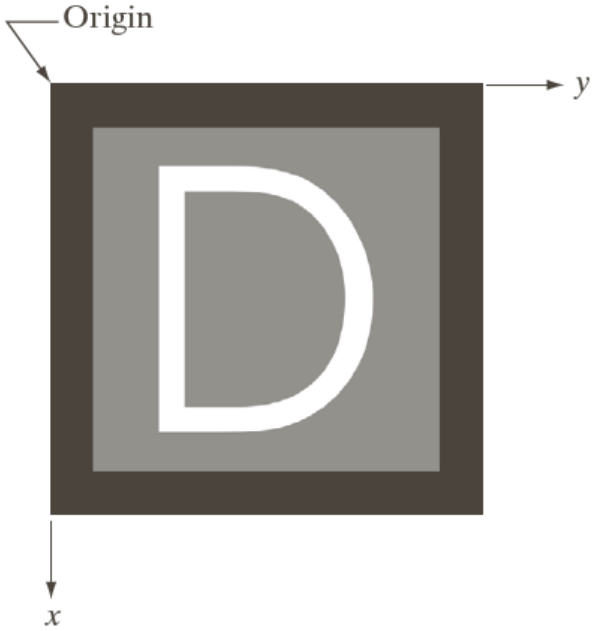


a	b
c	d

Generating a digital image. (a) Continuous image. (b) A scaling line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) sampling and quantization. (d) Digital scan line.

Example





Origin

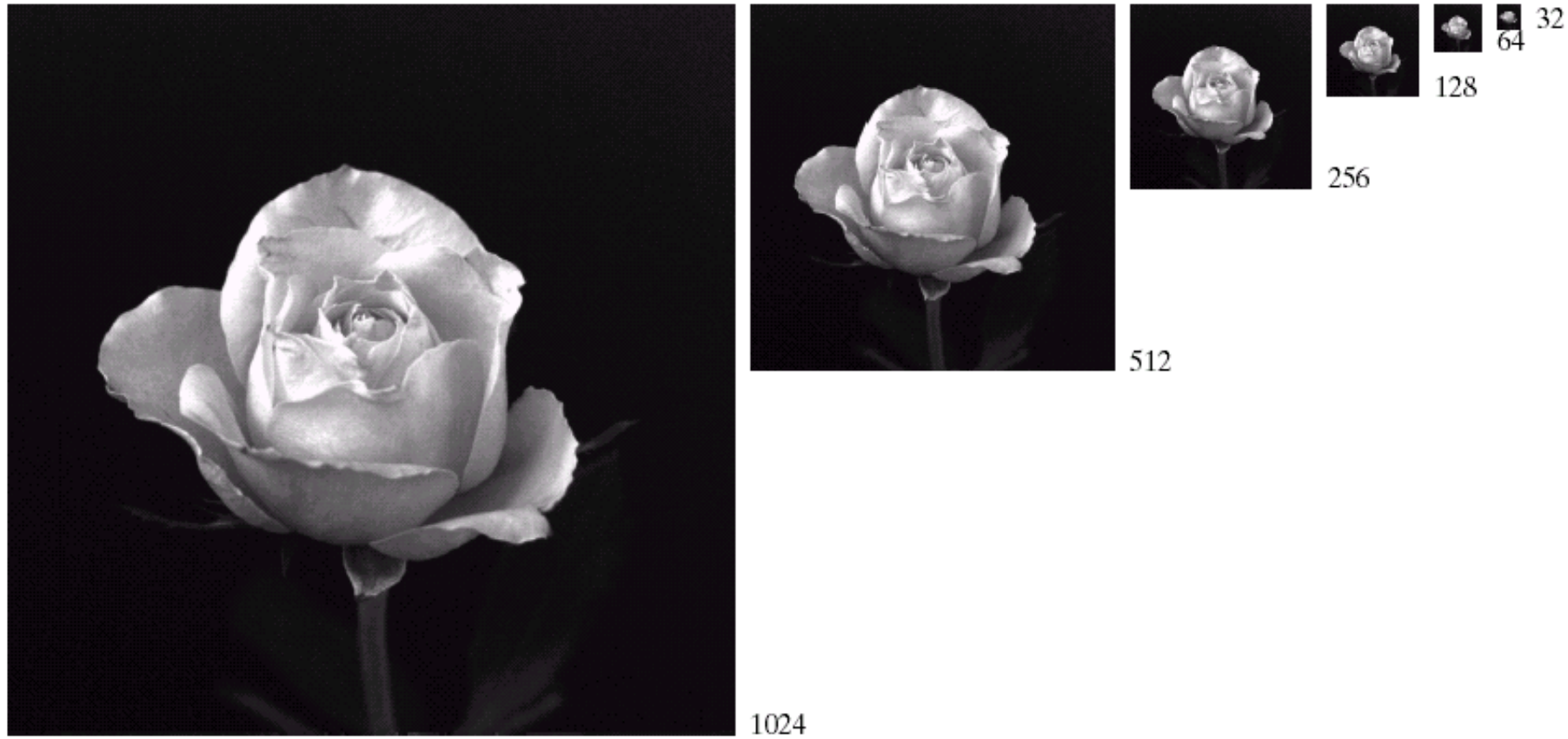
$$
\begin{array}{ccccccccccccccccccccccc}
0&0&0&0&0&0&0&\cdots&0&0&0&0&0&0&0\\
0&0&0&0&0&0&&&&0&0&0&0&0&0\\
0&0&0&0&0&&&&&&0&0&0&0&0\\
0&0&0&0&&\vdots&&&&&&0&0&0&0\\
0&0&0&&\cdots&.5&.5&.5&\cdots&&&&&0&0&0\\
0&0&0&&&.5&.5&&&&&&&0&0&0\\
\vdots&&&&&.5&&\ddots&&&&&&&&\vdots\\
&&&&&\vdots&&1&1&1&\cdots&&&&\\
&&&&&&&1&1&&&&&&\\
0&0&0&&&&&1&&\ddots&&&&0&0&0\\
0&0&0&&&&&\vdots&&&&&&0&0&0\\
0&0&0&0&&&&&&&&&0&0&0&0\\
0&0&0&0&0&&&&&&&0&0&0&0&0\\
0&0&0&0&0&0&&&&&0&0&0&0&0&0\\
0&0&0&0&0&0&0&\cdots&0&0&0&0&0&0&0
\end{array}
$$

Figure: (a) Image plotted as surface, (b) Image displayed as a visual intensity array, (c) Image shown as a 2-D numerical array (0, .5, 1 represent black, gray, and white respectively)

Spatial and Intensity Resolution

<u>Spatial Resolution</u>

- It refers to the smallest discernable detail in an image.
- The spatial resolution of an image is determined by how sampling was carried out.
- It is measured in dots per inch (DPI) or pixels.


1024
512
256
128
64
32

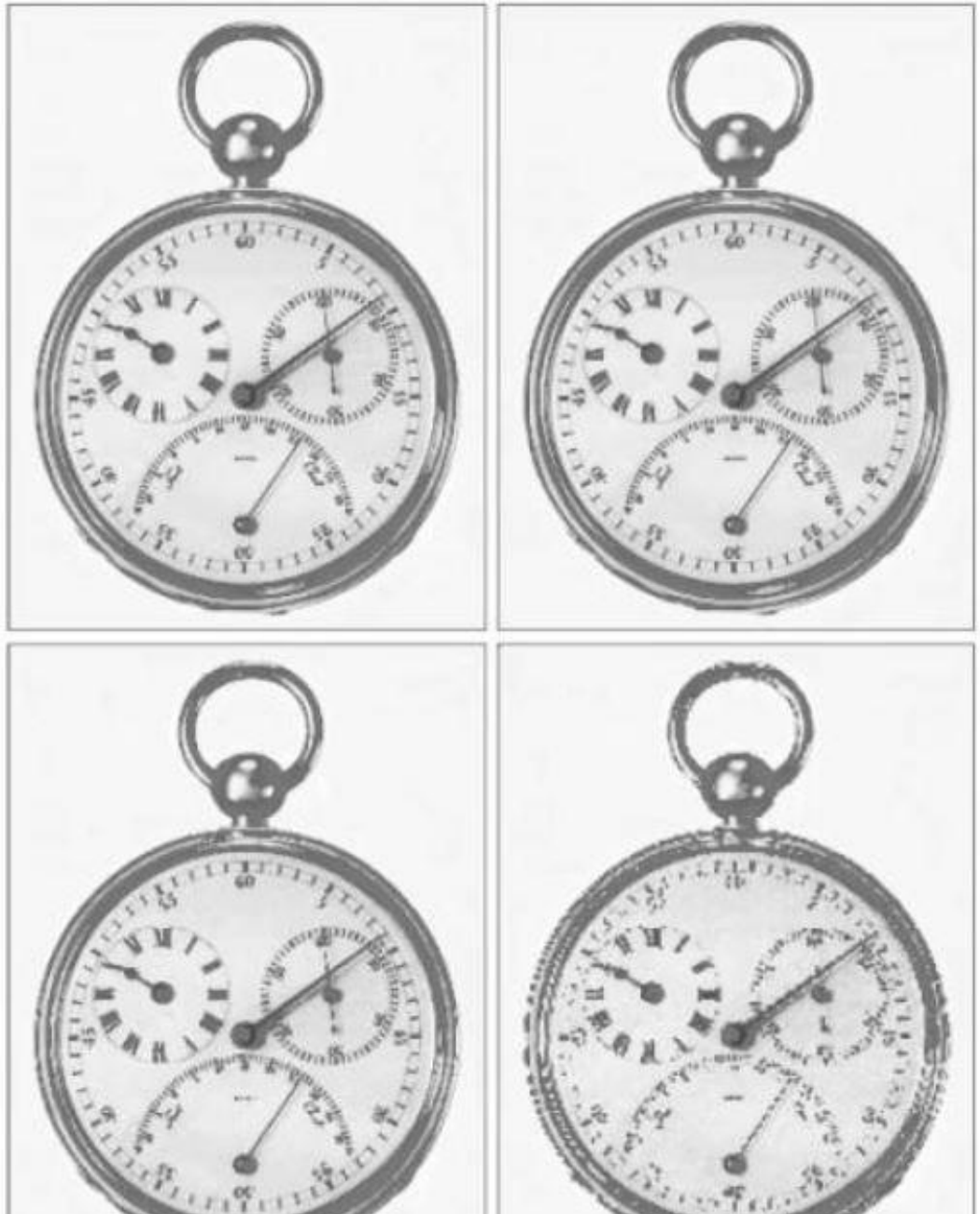
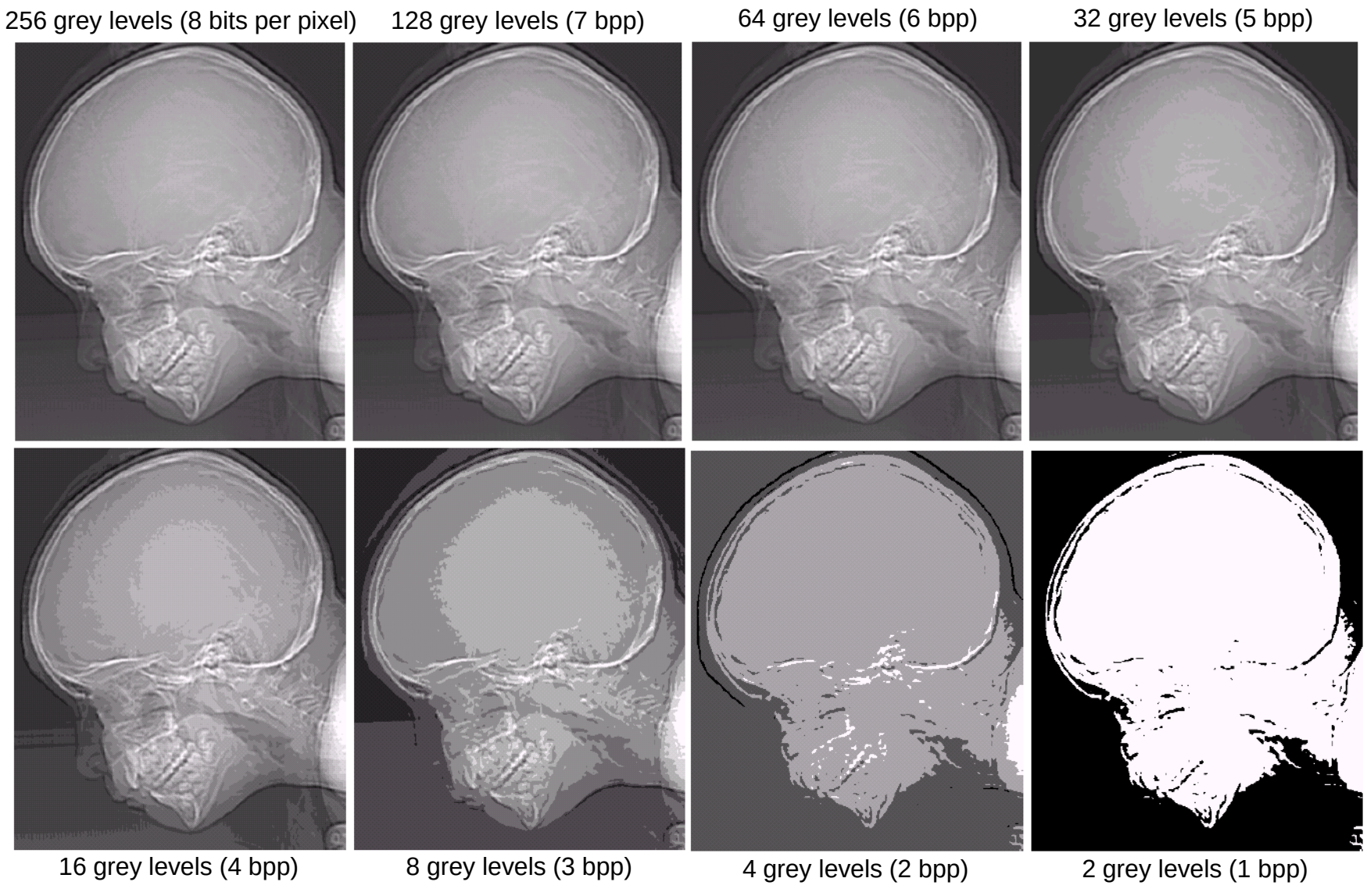


Figure: Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi.

Intensity Level Resolution

- Intensity Level Resolution refers to the number of intensity levels used to represent the image.
- The more intensity levels used, the finer the level of details discernable in an image.
- Intensity level resolution is usually given in terms of the number of bits used to stored each intensity level.

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

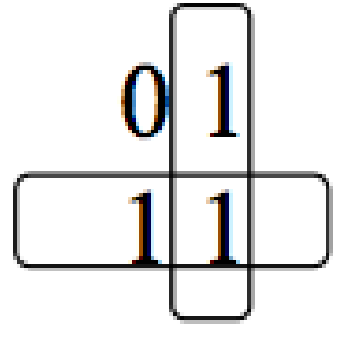

256 grey levels (8 bits per pixel)
128 grey levels (7 bpp)
64 grey levels (6 bpp)
32 grey levels (5 bpp)
16 grey levels (4 bpp)
8 grey levels (3 bpp)
4 grey levels (2 bpp)
2 grey levels (1 bpp)

Some Basic Relationships Between Pixels

- An image is denoted by: f(x, y)
- Lowercase letters (e.g. *p*, *q*) will denote individual pixels
- A subset of f(x, y) is denoted by *S*
- Neighbours of a pixel:
  - A pixel *p* at (x, y) has 4 horizontal and vertical neighbours at
    - (x+1,y),(x-1,y),(x,y+1)and(x,y-1) is called the *4-neighbors of p* and denoted by $N_4(p)$
  - A pixel *p* at (x,y) has 4 diagonal neighbours at (x+1,y+1),(x+1,y-1),(x-1,y+1) and (x-1,y-1) and is called the *diagonal–neighbors of p* and is denoted by $N_D(p)$
  - The *4-neighbors* and the *diagonal-neighbors* of *p* are called the *8-neighbors of p* and is denoted by $N_8(p)$

Connectivity Between Pixels

- Connectivity is an important concept in establishing boundaries of object and components of regions in an image
- When are two pixels connected?
  - If they are adjacent in some sense (say they are 4-neighbors)
  - and, if their gray levels satisfy a specified criterion of similarity (say they are equal)
- Example: given a binary image (e.g. gray scale = [0,1]), two pixels may be 4-neighbors but are not considered connected unless they have the same value



Connectivity between pixels

- Let *V* be the set of values used to determine connectivity
  - For example, in a binary image, *V*={1} for the connectivity of pixels with a value of 1
  - In a gray scale image, for the connectivity of pixels with a range of intensity values from 0 to 255 i.e. set V could be any subset of these 256 values.

we consider three types of connectivity

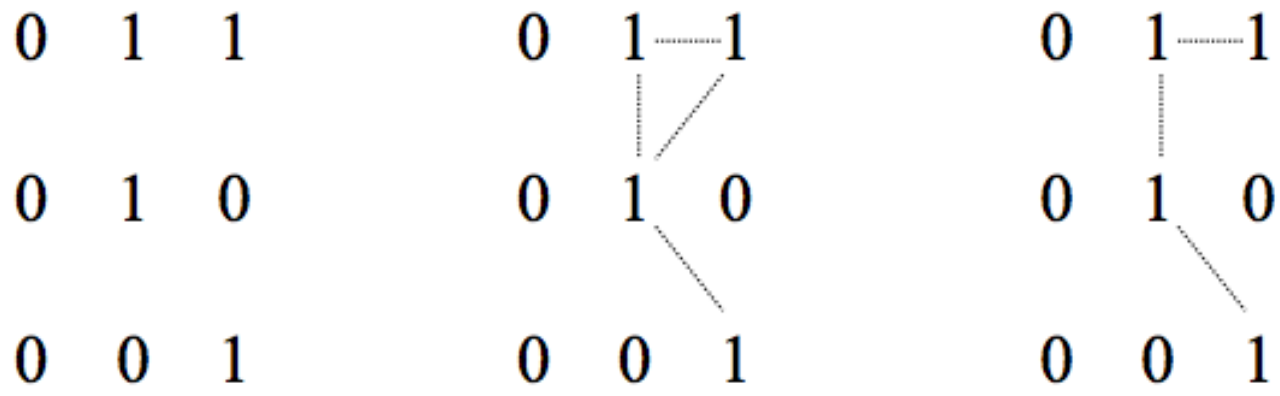
- *4-connectivity*: Two pixels *p* and *q* with values from *V* are *4-connected* if *q* is in the set $N_4(p)$
- *8-connectivity*: Two pixels *p* and *q* with values from *V* are *8-connected* if *q* is in the set $N_8(p)$
- *m-connectivity (mixed)*: Two pixels *p* and *q* with values from *V* are *m-connected* if

  i) *q* is in the set $N_4(p)$, *or*

  ii) *q* is in the set $N_D(p)$ *and* the set $N_4(p) \cap N_4(q)$ is empty (This is the set of pixels that are 4-neighbors of *p* and *q* and whose values are from *V*)

Connectivity between pixels

- Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise with 8-adjacency is used.
-



An arrangement of pixels

8-connectivity of the pixels V={1}

m-connectivity of the pixels V={1}

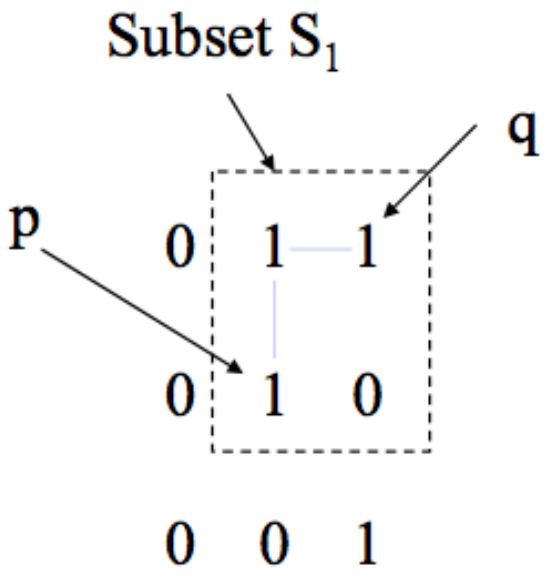
Pixel Adjacencies and Paths

- A path from pixel *p* at (*x,y*) to pixel *q* at (*s,t*) is a sequence of distinct pixels with coordinates

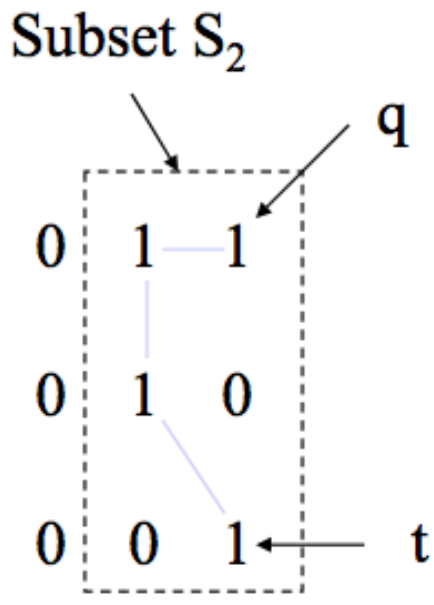
$$(x_0,y_0), (x_1,y_1),....., (x_n,y_n)$$

  - Where $(x_0,y_0)=(x,y)$ and $(x_n,y_n)=(s,t)$ and
  - (x_i,y_i) is adjacent to (x_{i-1},y_{i-1}) for $1<= i <= n$
  - n is the *length* of the path
- We can define 4-, 8-, or m- paths depending on the type of adjacency specified.
- Let S represent a subset of pixels in an image. Two pixels *p* and *q* are said to be connected in *S*, if there is a path from *p* to *q* consisting entirely of pixels in *S.*

Examples of Paths



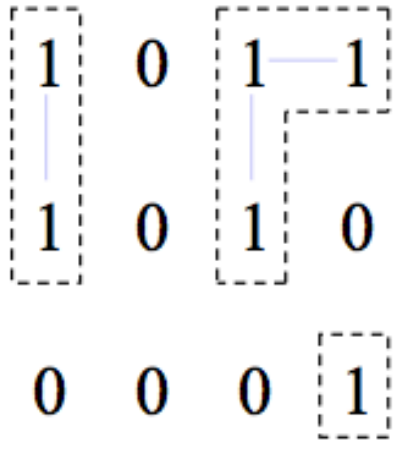
A 4-connected path from p to q (n=2). p and q are connected in S_1



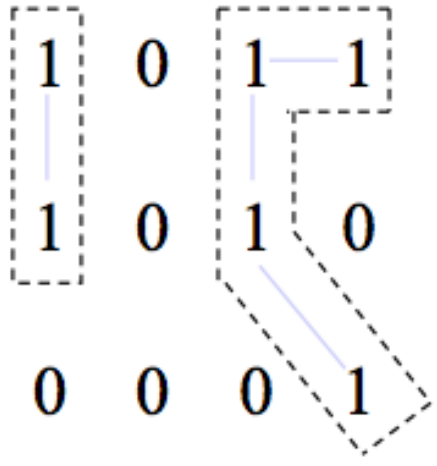
An m-connected path from t to q (n=3). t and q are connected in S_2

Connected Components

- For any pixel *p* in s, the set of pixels connected to *p* from a connected components of *S*.
- Distinct connected components of *S* are said to be *disjoint*.



3 4-connected components of S



2 m-connected components of S

Distance Measures

For given pixels *p*, *q*, and *z* at (*x*,*y*), (*s*,*t*) and (*u*,*v*) respectively, *D* is a *distance function* (or *metric*) if:

- $D(p, q) \geq 0$ ($D(p, q)=0$ iff $p=q$),
- $D(p, q) = D(q, p)$, and
- $D(p, z) \leq D(p, q) + D(q, z)$.

- The *Euclidean distance* between *p* and *q* is given by:

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

- The pixels having distance less than or equal to some value *r* from (x, y) are the points contained in a disk of radius *r* centred at (x, y).

Distance Measures

- The D_4 distance (also called the *city block distance*) between *p* and *q* is given by:

$$D_4(p,q) = |x-s| + |y-t|$$

- The pixels having a D_4 distance less than some *r* from (x,y) form a diamond centred at (x,y)
- Example: pixels where $D_4 \leq 2$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

Note: Pixels with D_4=1 are the 4-neighbors of (x,y)

Distance Measures

- The D_8 distance (also called the *chessboard distance*) between *p* and *q* is given by:

- The pixels having a D_8 distance less than some *r* from (x,y) form a square centred at (x, y)
- Example: pixels where $D_8 \leq 2$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Note: Pixels with $D_8=1$ are the 8-neighbors of (x,y)

Distance Measures and Connectivity

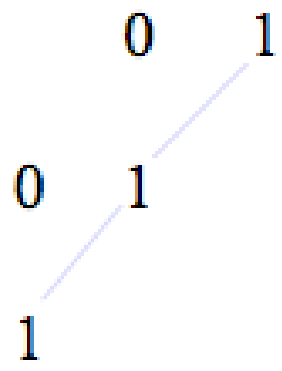
- The D_4 distance between two points *p* and *q* is the shortest 4-path between the two points.
- The D_8 distance between two points *p* and *q* is the shortest 8-path between the two points.
- D_4 and D_8 may be considered indepedent, regardless of whether a connected path exists between them, because the definition of these distances involves only the pixel coordinates.
- For *m-connectivity*, the value of the distance (the length of the path) between two points depends on the values of the pixels along the path.

Distance Measure and m-connectivity

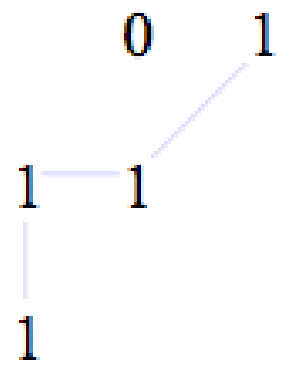
$$\begin{array}{ccc} & p_3 & p_4 \\ p_1 & p_2 & \\ p & & \end{array}$$

- Consider the given arrangement of pixels and assume
  - p, p_2 and p_4 =1
  - p_1 and p_3 can be 0 or 1
- If V={1} and p_1 and p_3 are 0, the m-distance (p, p_4) is 2. If either p_1 or p_3 are 1, the m-distance (p, p_4) is 3. If p_1 and p_3 are 1, the m-distance (p, p_4) is 4.

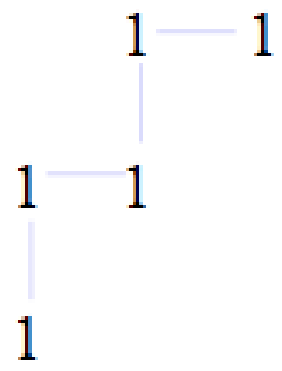
M-Connectivity Example



m-distance(p,p_4)=2



m-distance(p,p_4)=3



m-distance(p,p_4)=4